Operational Information and Decision Value in Rolling Airline Recovery: From Recovery Screening to Local Recovery

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ARTICLE INFO

Keywords:
airline recovery
operational information
recovery screening
decision support
uncertainty representation
rolling decision-making

ABSTRACT

Observed delay is timely and interpretable for airline disruption recovery, but the value of richer operating information depends on whether it changes the recovery decision being supported. This study examines how the incremental decision value of progressively revealed aircraft-linked operating information varies across information representations and recovery decisions. We develop a state-consequence-decision framework linking capacity-constrained screening with local-recovery targeting, and compare Delay-Consequence, Current-History, and Point-Marginal-Joint representations. The data infrastructure spans U.S. airline operations from 2017 to 2022, while primary held-out decision evidence comes from a frozen 2019 chronological evaluation. Delay remains informative at the population level, yet global agreement does not guarantee decision substitutability at the active screening boundary: under the nominal 10% review capacity, five TURN shortlist positions change and $L^{\mathrm{att}} = 0.0724$. History produces limited Stage-I loss but much larger Stage-II loss (0.0186 versus 0.9070). Point compression also causes substantial decision loss, whereas restoring marginal uncertainty recovers most of the observed Point-to-Joint difference; additional Joint alignment adds relatively little. Operational information value is therefore decision specific and boundary dependent, rather than proportional to representational richness.

1. Introduction

Airline disruption recovery is a continuous operational allocation problem. After disruptions occur, airline operations control centers (AOCCs) must determine which affected flight chains deserve immediate attention, which disruptions require more detailed diagnosis, and where recovery effort should ultimately be deployed under limited time, analytical capacity, and operational resources. Effective recovery support therefore requires more than identifying disruption severity: it must translate progressively available operational information into signals that are matched to the recovery decision being made.

Delay provides a natural baseline for this task. It is interpretable and continuously updated, making it well suited to rapid operational screening. In the decision-time comparisons of this study, the scalar delay priority is the expected successor take-off delay conditional on information then available, rather than the delay eventually observed after the flight is completed. Yet disruption consequences are not determined by delay magnitude alone. As aircraft-linked flight chains evolve, realized predecessor operations, turnaround conditions, and successor states progressively clarify the operating situation. Passenger, route, and downstream exposure references already available at the decision time may also distinguish the consequences of similarly delayed chains. Existing airline recovery research has documented the multidimensional nature of flight, passenger, and network consequences (Bratu and Barnhart, 2006; Arikan, Gurel and Akturk, 2016; Clausen, Larsen, Larsen and Rezanova, 2010; Hassan, Santos and Vink, 2021). The relevant question is therefore not simply whether richer operating information describes more detail than delay, but how much incremental decision value those additional state and consequence distinctions provide.

Information value also depends on the recovery decision being supported. Stage-I screening allocates limited review attention across disrupted flight chains, whereas subsequent local recovery evaluates whether a shortlisted chain should receive additional recovery effort and which recovery target should be selected. Existing staged-recovery studies similarly distinguish prioritization, affected-flight identification, and subsequent recovery or schedule adjustment (Pei, He, Fan and Zhang, 2021; Ji, Gao, Zhang and Li, 2021; Zhao, Bard and Bickel, 2023; Zhong, Lian, Zhou and Niu, 2024). In our framework, screening is based on relative baseline consequence exposure, while local recovery weighs the remaining recoverable opportunity against the effort required to pursue it. A disruption may consequently warrant high screening priority without retaining a beneficial local-recovery opportunity, and information that leaves the shortlist

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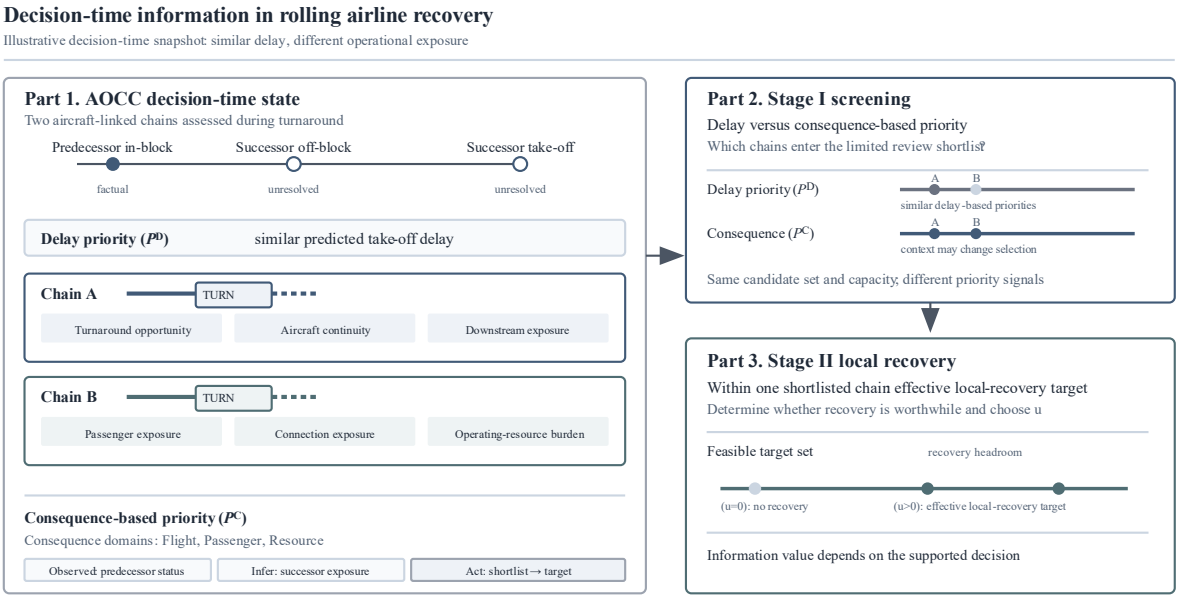


Figure 1: Decision-time information and linked recovery decisions in an aircraft-linked disruption setting. Progressively revealed operating information can distinguish flight chains beyond observed delay, first affecting capacity-constrained screening across chains and subsequently informing effective local-recovery targeting within shortlisted actionable chains.

unchanged may still alter the downstream recovery target. These distinct comparisons make the value of operating information specific to the decision being supported. Figure 1 illustrates this staged structure.

The representation of operational information provides a second source of variation in decision value. Temporal information can retain either the current operating state alone or the trajectory through which that state was reached. Likewise, unresolved future states can be compressed into a deterministic estimate or represented in forms that preserve different aspects of uncertainty. Existing research on delay propagation and dynamic recovery has emphasized temporal evolution, aircraft rotations, and historical operating information (Fleurquin, Ramasco and Eguiluz, 2013; Li and Jing, 2021; Erdem and Bilgiç, 2024), while probabilistic approaches retain multiple possible future realizations rather than collapsing uncertainty to a single point estimate (Woo and Moon, 2021; Zoutendijk and Mitici, 2021; Ogunsina, Papamichalis and DeLaurentis, 2022; Dalmau, De Falco, Spak and Varela, 2024). We examine these distinctions through Current versus History information and Point, Marginal, and Joint representations of unresolved aircraft-linked states.

Richer representation need not produce proportional decision refinement: incremental value arises when the additional distinctions retained by a representation alter the comparison relevant to the supported decision. For screening, this may involve the relative positions of flight chains near a fixed review-budget boundary. For local recovery, it may involve the consequence–effort trade-off between feasible recovery targets. Managerial objectives, nonlinear consequence thresholds, dependence among operating states, and remaining recovery headroom can affect whether a representational difference changes the preferred target. Accordingly, we evaluate not only whether information representations lead to different choices but also the model-relative loss of those choices under a common reference criterion.

Against this background, this study asks the following overarching research question:

> *In rolling airline recovery, how much incremental decision value does aircraft-linked operating information beyond delay provide, and how does this value vary with progressively revealed operating states, information representations, and the recovery decisions being supported?*

The question has two linked dimensions. The first concerns operating information and its representation: what distinctions are added by realized milestones, historical trajectories, and alternative representations of unresolved

uncertainty beyond a compact delay signal. The second concerns the supported recovery decision: whether these distinctions affect cross-chain screening, within-chain local-recovery targeting, or both. Managerial objectives, decision boundaries, and operating conditions are examined as conditions that may explain variation in incremental decision value. In this sense, decision-specific information sufficiency is an interpretation of the comparisons—whether a more compact representation preserves the distinctions needed for a particular choice—rather than a presumption that richer representations must always be preferable.

To investigate this question, we develop a unified state–consequence–decision framework with two linked recovery stages. Stage I considers fixed-budget screening, where limited review attention is allocated across disrupted aircraft-linked flight chains using expected-delay or multidimensional-consequence priorities. A delay-plus-exposure comparator helps distinguish the role of historical exposure references from that of a scalar delay signal. Stage II selects an effective local-recovery target for a chain with valid recovery support under a common multidimensional consequence–effort objective. Controlled Current–History and Point–Marginal–Joint comparisons assess how temporal information and uncertainty representation affect these decisions. Additional comparisons of managerial objectives, state-dependence mechanisms, and decision boundaries examine when the information differences become relevant to local recovery. All comparisons assess model-relative decision value under defined reference criteria, not observed causal effects of recovery actions.

The empirical design covers six independent operating years in U.S. aviation from 2017 to 2022, using consistent aircraft-linked chain definitions, recovery tasks, and within-year chronological held-out evaluation. This window includes pre-pandemic operations (2017–2019), the COVID-19 disruption period (2020), and the subsequent operational recovery period (2021–2022). The frozen 2019 evaluation remains a traceable original benchmark, while the six year-specific evaluations form the intended evidence base for the main comparisons and their variation across operating conditions.

The study makes three contributions. First, it moves the comparison of delay and multidimensional consequences beyond population-level association to operational decision substitutability. By linking screening choices to fixed review budgets, consequence coverage, and the active selection boundary, and by adding a delay-plus-exposure comparator, the analysis evaluates the incremental value of consequence distinctions for allocating limited managerial attention. Second, it distinguishes baseline consequence priority from remaining recoverable opportunity while examining the decision effects of temporal history and uncertainty representation within the same aircraft-linked operating process. The Current–History and Point–Marginal–Joint contrasts allow incremental information value to be evaluated across both screening and local-recovery tasks under common reference criteria, rather than inferred from predictive accuracy alone. Third, it relates the decision-boundary interpretation of incremental information value to managerial objectives, state-dependent recovery opportunities, and operating conditions. Through controlled objective and dependence comparisons together with year-specific evaluations, the analysis examines when simpler representations preserve the distinctions needed for recovery decisions and when additional information can change those choices.

The remainder of the paper is organized as follows. Section 2 reviews the related literature and positions the study around operating consequences, rolling information, and decision value. Section 3 defines the aircraft-linked operating state, multidimensional consequences, two-stage recovery decisions, and the reference criteria for incremental decision value and information sufficiency. Section 4 presents the six-year empirical design, state reconstruction, consequence construction, decision implementation, and controlled comparisons. Section 5 reports the evidence on screening, recoverable opportunity, information representations, managerial objectives, and operating-stage and regime differences. Section 6 discusses decision-specific information value, implications for AOCC decision support, and applicability. Section 7 concludes.

2. Related Literature and Research Positioning

2.1. Airline Recovery Decisions and Multidimensional Operational Consequences

Airline operations are closely interconnected, and a disruption to an individual flight can affect not only its own schedule execution but also subsequent operations through aircraft turnaround, crew connections, passenger itineraries, and airport resource use. Recovery decisions therefore require coordination across interdependent operational elements rather than treating disruptions solely as deviations from scheduled flight times. Clausen et al. (2010) review aircraft, crew, and passenger recovery and highlight the interdependencies created by schedules, resource connections, and recovery actions. By incorporating passenger delays and itinerary disruptions into flight recovery, Bratu and Barnhart

(2006) demonstrate why flight-level operating costs alone may not adequately represent the consequences of a recovery plan. Hassan et al. (2021) further review developments in integrated recovery and identify continuing gaps between optimization models and airline operating practice. Together, these studies indicate that delay is an important indicator of disruption, but its temporal magnitude captures only part of the wider operational consequences.

This multidimensionality also means that flights with similar delays may have substantially different operational implications. Subsequent aircraft assignments, the number and connections of affected passengers, and resource pressures under different airport and operating conditions can all shape the significance of a disruption. The integrated aircraft–passenger recovery formulation of Arikan et al. (2016) likewise illustrates the relationship between operating adjustments and passenger outcomes. Yet assessing multiple consequences raises a further question: how should outcomes measured in different units, with different managerial meanings and relative importance, be compared? Multiattribute decision analysis provides a basis for making such trade-offs explicit. Keeney and Raiffa (1976) examine preferences and value trade-offs across decision objectives, while OECD, European Union and European Commission, Joint Research Centre (2008) emphasize normalization, weighting, and sensitivity analysis in constructing composite indicators. These approaches support transparent multidimensional evaluation, but the weights and aggregation rules still require a defensible managerial interpretation. Standardization alone does not turn a composite measure into an observed economic cost or a decision maker's elicited utility.

Beyond evaluating consequences, airline recovery involves different levels of managerial decision making. During operations, controllers assess disruption importance and current operating conditions to determine which problems deserve attention and whether further adjustments are warranted. Pei et al. (2021) link real-time flight assessment with subsequent recovery recommendations in a decision-support approach for irregular operations. Ji et al. (2021) incorporate flight priorities and airport-capacity constraints into recovery planning, reflecting trade-offs between operational objectives and constrained resources. Under uncertainty, Zhao et al. (2023) distinguish the identification of affected flights from subsequent schedule adjustments, illustrating the staged nature of information revelation and recovery planning. Airline recovery is therefore not a single task governed by one uniform comparison, but involves disruption identification, priority assessment, evaluation of recovery opportunities, and operational adjustment.

These tasks are related without having identical decision objects. Consequence severity describes the potential impact of a disruption on the operating system, whereas recoverable value also depends on the adjustment opportunities that remain, the available alternatives, and the effort required to improve outcomes. A highly consequential disruption may warrant early attention without offering much scope for additional local recovery; conversely, a less severe disruption may allow a more readily attainable improvement. The staged-recovery literature provides a basis for recognizing this distinction and, in turn, suggests that different tasks need not rely on operating information in the same way. An accurate assessment of disruption severity does not, by itself, constitute an equally informative assessment of recovery opportunity.

2.2. Rolling Operational Information and Decision-Based Evaluation

Airline recovery decisions are commonly made before the operating process has finished and while future states remain uncertain. As flights progress, aircraft arrivals, turnaround activities, subsequent flight execution, and airport conditions are continually updated, with previously unresolved events becoming observed facts. Vink, Santos, Verhagen, Medeiros and Filho (2020) incorporate evolving disruptions and operational conditions into a dynamically updated aircraft-recovery framework, while Lee, Marla and Jacquillat (2020) study recovery decisions under airport operating uncertainty. The treatment of disruption duration and information revelation in Zhao et al. (2023) likewise underscores that the information available to decision makers changes over time. Operational information in airline recovery thus concerns not only forecasting future states, but also using newly available evidence as operations unfold.

As the operating state evolves, its current description and the history leading to it may offer different bases for assessment. The current state summarizes conditions observable or inferable at a given time, whereas the preceding trajectory may retain information about how delays developed, propagated, and changed. Fleurquin et al. (2013) study systemic delay propagation through interactions in the airline network, Li and Jing (2021) characterize delay propagation in the air-traffic network, and Erdem and Bilgiç (2024) examine delay estimation and evolution in daily operations. These studies establish the temporal interconnectedness of airline operations, but do not imply that historical trajectories necessarily add decision value beyond the current state. The usefulness of history depends on how much of the relevant past is already summarized by current information. When the current representation is sufficient for the subsequent assessment, retaining the trajectory may add complexity without a corresponding benefit; when it omits relevant dynamics, history may provide additional insight.

More complete state information, however, does not necessarily translate into better decisions. Statistical prediction is often assessed through estimation error, probability calibration, or uncertainty representation, whereas managerial decisions depend on the consequences of the alternatives selected. Blackwell (1953) establish a foundational comparison of information structures in statistical decision theory, and Howard (1966) emphasize that the value of information cannot be separated from the consequences of uncertain events. Avriel and Williams (1970) relate information value to stochastic programming. This decision-oriented perspective has also informed more recent predictive modeling: Donti, Amos and Kolter (2017) incorporate downstream optimization tasks into learning, while Elmachtoub and Grigas (2022) show why predictive-error improvements need not translate directly into improved optimization decisions. Despite differences in setting and formulation, these contributions distinguish the quality of information from its performance in a particular decision task.

This distinction is especially pertinent to airline recovery because different tasks require different comparisons. Disruption screening concerns which flights deserve priority relative to other candidates; local recovery concerns whether, and how far, a particular flight should be adjusted given the consequences and effort associated with alternative choices. The former depends on cross-flight comparisons, whereas the latter depends on comparisons among recovery options within a flight. Additional information may change an operating-state assessment without changing either decision, and a changed decision may carry only a small difference in the relevant objective. The managerial value of information consequently depends both on the operating-state distinctions it preserves and on their relevance to the particular decision and evaluation criterion.

2.3. Decision Objectives and Distributional Information Requirements

When future operations remain uncertain, decision support also depends on how that uncertainty is represented. A point prediction offers a compact summary but does not describe the probabilities of alternative realizations. A probability distribution retains this variation, while a joint distribution can additionally represent dependence among uncertain quantities that jointly influence operating consequences. Scenario-based approaches already illustrate the importance of uncertainty in airline operations. Woo and Moon (2021) use stochastic scenarios for flight rescheduling, Ogunsina et al. (2022) model uncertainty and its propagation using a relational dynamic Bayesian network, and Zoutendijk and Mitici (2021) apply probabilistic delay prediction to subsequent airport decisions. In the broader decision-analysis literature, Clemen and Reilly (1999) explain how marginal distributions and copulas can be used to construct joint probability models, distinguishing uncertainty about individual variables from dependence between them.

A more complete probability representation need not, however, yield a comparable improvement in decision performance. Joint distributions describe the co-occurrence of uncertain states, but their usefulness depends on the role that such joint realizations play in the decision problem. Agrawal, Ding, Saberi and Ye (2012) examine the cost of neglecting correlations in stochastic optimization and show that it depends on problem structure: the resulting loss can be controlled in some classes of problems but can be substantial in others. Thus, the importance of dependence information cannot be inferred solely from the existence of statistical correlation. The amount of structure preserved in a representation and the extent to which that structure improves a decision are different considerations.

The decision objective itself also determines which probability features enter the evaluation of an alternative. For an expected objective that is additive across uncertain variables, preserving the relevant marginal distributions may suffice to evaluate each fixed alternative. By contrast, when consequences depend on joint realizations or the effect of an adjustment is constrained by relationships among uncertain states, dependence can affect the expected performance of different alternatives. Clemen and Reilly (1999) examine dependence modeling and sensitivity to correlations in decision analysis, while Wang and Dyer (2012) develop a copula-based approach to representing dependent uncertainties in multivariate decision trees. Together with the problem-dependent costs of ignoring correlation studied by Agrawal et al. (2012), these contributions indicate why the relevance of dependence must be assessed in relation to how an objective translates uncertain states into consequences. Different managerial objectives can therefore place different demands on the same underlying probability representation.

Even when dependence changes the evaluation of possible outcomes, it need not change the final choice. Decisions involve comparisons among feasible alternatives, so what matters for selection is whether a change in probability representation alters their relative performance. A representation can change expected objective values while preserving the preferred alternative; conversely, even a modest relative change can reverse the choice when alternatives are closely matched. The decision-consequence perspective of Howard (1966), together with the downstream objective-based evaluation developed by Elmachtoub and Grigas (2022), provides a basis for distinguishing informational refinement

from decision improvement. The managerial value of distributional information must therefore be considered in relation to probability structure, decision objectives, and the comparisons among alternatives, rather than judged by representational completeness alone.

2.4. Research Gap

Existing research provides important foundations for uncertainty modeling in airline recovery and for decision-based information evaluation. The connection between these perspectives in specific airline operating decisions, however, warrants further empirical examination.

For the same operating process, there remains a need for systematic evidence on when simpler operating signals or probability representations are sufficient for different recovery tasks. In particular, it is not yet well established under what conditions a delay-based signal preserves the recovery priorities supported by multidimensional consequences, when current-state information or marginal uncertainty can substitute for historical trajectories or joint dependence, and how the value of these distinctions varies with managerial objectives and operating conditions. The central issue is not merely whether a richer representation improves the description of the operating state, but whether the distinctions it preserves change the comparison among recovery alternatives and the consequences of the resulting decisions.

Against this background, the present study examines the roles of operating information and its representations in recovery screening and local-recovery decisions under rolling airline operations. It further considers how decision objectives and operating conditions shape their decision relevance, with the aim of identifying when simpler information supports the same decisions and when richer operating information offers incremental value.

3. Decision Framework

This section develops a decision framework for evaluating the sufficiency of operational information in rolling airline recovery. It links aircraft-specific operating states and multidimensional consequences to two connected decisions: identifying disrupted flight chains for further review under limited screening capacity, and selecting a local recovery target for a shortlisted, actionable chain. The two decisions share operating-state and consequence definitions but differ in their decision objects and optimality criteria. By holding the decision-time evidence boundary and the relevant managerial objective fixed, the framework assesses whether alternative information representations change the selected decisions and the consequences of those changes under a common reference criterion. Figure 2 summarizes these relationships. Throughout, i identifies a predecessor–successor flight chain, t a decision epoch, g its operating stage, s a conditional scenario, and u a local recovery target. Where a canonical decision epoch is fixed for a chain–stage case, we suppress t for readability.

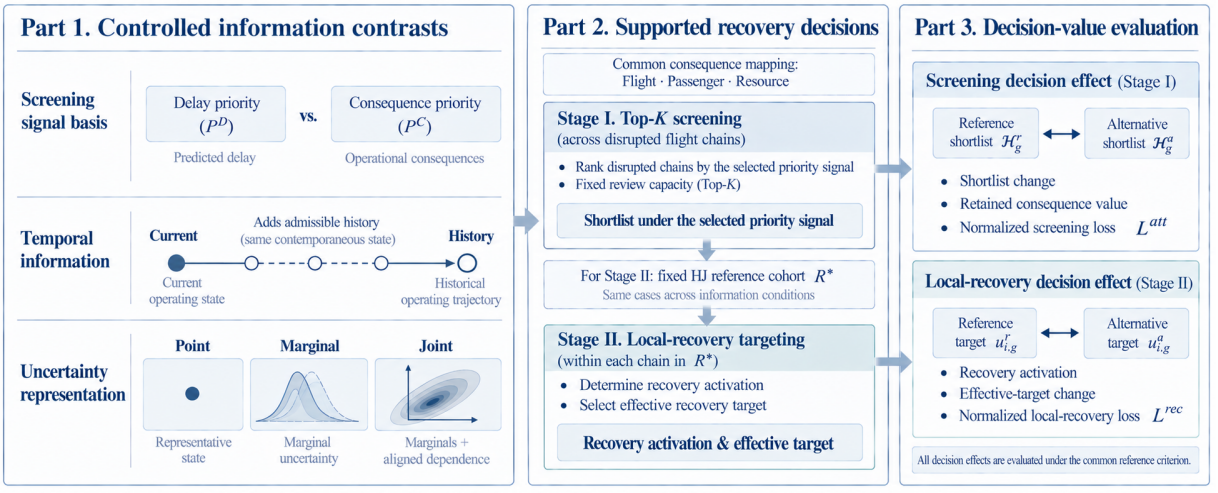


Figure 2: Controlled operating-information contrasts and two recovery decisions. Delay–Consequence changes the Stage-I screening signal; Current–History changes temporal representation; and Point–Marginal–Joint changes the representation of unresolved uncertainty. Stage-specific screening and within-chain recovery targeting are assessed against their respective reference decision criteria.

3.1. Aircraft-Linked Recovery Setting and Decision Contexts

We consider day-of-operation decisions in an airline operations control center (AOCC) setting. The operating unit is an aircraft-linked predecessor–successor flight chain,

$$f_i^- \longrightarrow f_i^+.$$

Following the predecessor's actual in-block event, the aircraft undergoes ground turnaround before operating the successor flight. A predecessor delay does not necessarily propagate to the successor by the same amount. Turnaround conditions, subsequent assignments, passenger exposure, and local operating burden all affect how delay is converted into downstream consequences. Flight chains with similar observed delays can therefore differ both in their baseline operational consequences and in their remaining recovery opportunities.

3.1.1. Operating stages

Four operating stages describe the progressive realization of aircraft-linked milestones. *PRE* precedes the predecessor's actual in-block event, when its arrival and subsequent operating states may remain unresolved. *TURN* begins after the predecessor has arrived but before the successor's actual off-block event. *TAXI* follows the successor's off-block event while taxi-out and take-off remain to be completed. *COMP* denotes completion of the focal operating process. As operations progress, previously uncertain milestones become factual and the unresolved state requiring prediction changes accordingly. Local off-block recovery is evaluated only at

$$\mathcal{G}_A = \{\text{PRE}, \text{TURN}\}.$$

These operating stages govern information availability and the remaining state uncertainty. They must not be confused with Stage I and Stage II, which designate *decision layers*, not operating milestones.

3.1.2. Recovery decision contexts

Recovery requires decisions at different managerial levels. Screening first identifies disruptions with potentially important operational consequences; detailed local assessment then considers whether an additional adjustment is worthwhile. This separation is consistent with the established distinction between disruption identification and recovery adjustment in airline recovery research (Pei et al., 2021; Zhao et al., 2023). *Stage I* selects chains for further diagnosis under a fixed review budget, using either delay-based or consequence-based priority. Let $z_{i,g} \in \{0, 1\}$ indicate whether chain i at operating stage g enters the shortlist. The principal empirical comparison is a *retrospective, stage-specific fixed-budget screening benchmark*: without verified concurrent candidate queues and actual AOCC review capacities, chains aggregated across different decision times are not interpreted as simultaneously competing for real review slots.

Stage II applies to shortlisted chains with valid local-recovery support and selects an effective recovery target $u_{i,g} \in \mathcal{U}_{i,g}$, including zero additional recovery. This variable specifies a modeled reduction in successor operating delay rather than an executable aircraft-swap, crew-assignment, gate-allocation, or other resource-dispatch instruction. Its realized effect remains constrained by the state-dependent recovery headroom. The decision layers share an operating-state and consequence mapping but optimize different objects: Stage I compares baseline consequence exposure across chains, whereas Stage II evaluates the achievable improvement and effort associated with different targets for one chain. Accordingly, a chain with high screening priority need not have high recoverable value. Neither layer-specific optimum establishes a globally optimal, resource-coordinated airline recovery policy. The review-budget and local-target assumptions are model specifications rather than observed airline policy rules.

3.2. Decision-Time Information and State Representations

Let $\mathcal{E}_{i,\le t}$ denote the admissible operating evidence available for chain i at or before decision epoch t. As milestones occur, unresolved operating events become realized facts. A separate modeling choice determines how the information that is already available is organized and represented. We therefore distinguish *information availability*, *temporal or distributional representation*, and the *decision objective*. Altering consequence-domain preferences changes the evaluation criterion rather than the amount of operating evidence; those preferences are introduced in Section 3.3.

The aircraft-linked state follows the operating sequence

$$T_{i,t}^{IB} \longrightarrow D_{i,t}^{OB} \longrightarrow D_{i,t}^{TX},$$

where T^{IB} is the predecessor in-block timing quantity, D^{OB} the successor off-block delay, and D^{TX} the successor's additional taxi-out delay. The arrows express operating sequence and the adopted conditional modeling order, not an

identified causal effect among the variables. All three primitives may be unresolved in PRE; the predecessor in-block event is factual in TURN; and both predecessor in-block and successor off-block are factual in TAXI. Only unresolved primitives receive probabilistic predictions, while realized milestones remain fixed. The temporal origin of T^{IB} and the native consequence units must be reconciled with the implementation before numerical claims are finalized.

3.2.1. Delay baseline and operating history

The primary scalar screening comparator is the *expected successor take-off delay* conditional on decision-time evidence, rather than the successor's already realized take-off delay:

$$P^D_{i,t} = \mathbb{E}\left[D^{TO}_{i,t} \mid \mathcal{E}_{i,\le t}\right], \qquad D^{TO}_{i,t} = D^{OB}_{i,t} + D^{TX}_{i,t}.$$

Here D^{TO} is the study-specific compound of off-block and additional taxi-out delays. The expected priority P^D must not be confused with actual take-off delay observed only after realization. A delay signal already observed at a decision epoch, when genuinely available, is a separate empirical comparator rather than a substitute for the stated baseline.

At a common decision epoch, the *Current* representation uses the contemporaneous operating-state summary and legally available frozen references. The *History* representation additionally uses the admissible trajectory through the same cutoff, encoded by an operator H_η:

$$\mathbf{h}_{i,t} = H_\eta\big(\mathcal{E}_{i,\le t}\big), \qquad \mathcal{I}^H_{i,t} = \mathcal{I}^C_{i,t} \cup \{\mathbf{h}_{i,t}\}.$$

Both representations describe the same operating process under the same information cutoff; neither may access a milestone realized after the decision. Current emphasizes the available state summary, whereas History attempts to retain aspects of delay development and state evolution not captured by that summary. The value of history is conditional: it can be redundant when the current state is sufficient for the downstream decision. Moreover, any measured difference between the implemented conditions may combine history access with changes in predictor capacity, estimation, and calibration. The comparison therefore first concerns the *implemented representations*; attribution to historical evidence itself requires the matched controls described in Section 4.

3.2.2. Point, Marginal, and Joint representations

For the operating primitives unresolved at the decision epoch, *Point* compresses uncertainty into one internally coherent representative state chosen by the frozen weighted-joint-medoid rule. It is not the vector of marginal means. *Marginal* preserves the weighted probability distribution of each unresolved primitive while replacing the original joint coupling. *Joint* preserves the original aligned, weighted scenarios and their cross-state dependence. Their conceptual sequence is

$$\text{Point} \longrightarrow \text{Marginal} \longrightarrow \text{Joint}.$$

This is a sequence of retained *representation features*, not successive acquisition of real observations and not an assumed ordering of decision value. Realized milestones remain identical across the representations.

To interpret a Marginal–Joint contrast specifically as the contribution of dependence, the two constructions must preserve the same *weighted* one-dimensional marginal probability measures. For every unresolved primitive S_j and measurable set B,

$$\Pr_{P_M}(S_j \in B) = \Pr_{P_J}(S_j \in B). \tag{1}$$

Preserving the coordinate values alone does not preserve their probability masses; permuting coordinate values while leaving unequal original scenario weights fixed generally violates Eq. (1). The equality is a required construction-and-audit property, not a claim that the previous implementation has satisfied it. Statistical marginal preservation also does not ensure that every recombined scenario respects the temporal order and resource restrictions of airline operations. The specific coupling algorithm, weighted-mass diagnostics, and operational-support checks are therefore reported separately in the empirical implementation and supplementary documentation.

3.3. State-Consequence-Decision Mapping

Each supported operating-state representation enters the same seven-component consequence mapping. The Flight domain covers aircraft continuity and turnaround pressure, successor execution delay, and downstream delay propagation. The Passenger domain covers passenger time exposure, itinerary-sensitive connection exposure, and prolonged-disruption service exposure. The Resource domain represents local operating burden through additional taxi-out delay. Write $\mathbf{C}_{i,g,s}(u) = (C_{i,g,s,k}(u))_{k=1}^{7}$ for the seven-dimensional consequence vector of chain i at operating stage g, scenario s, and local recovery target u. Several components are constructed exposure proxies rather than directly observed financial losses.

Because these components have different native units, they are converted using training-period reference scales into standardized, dimensionless consequence units (CU). The criterion aggregates them first within each domain and then across domains:

$$\begin{aligned} D_{i,g,s,d}(u) &= \sum_{k \in \mathcal{K}_d} \alpha_{d,k} Z_{i,g,s,k}(u), \\ \Phi_{\boldsymbol{\omega}}\big(\mathbf{C}_{i,g,s}(u)\big) &= \sum_{d \in \{F,P,R\}} \omega_d D_{i,g,s,d}(u). \end{aligned} \tag{2}$$

Here $Z_{i,g,s,k}(u)$ denotes a standardized consequence, $\mathcal{K}_d$ the components assigned to domain d, $\alpha_{d,k}$ their fixed within-domain weights, and $\boldsymbol{\omega} = (\omega_F, \omega_P, \omega_R)$ the normative weights across the Flight, Passenger, and Resource domains. Under the primary *Balanced* criterion, each domain receives one-third of the total weight; within Flight and Passenger, the three components receive equal within-domain shares. Thus Balanced means *equal domain weights*, not equal weights for all seven components. Prespecified Flight-, Passenger-, and Resource-emphasis criteria change the managerial objective without changing the available operating evidence. Standardization addresses measurement scales, whereas weights express trade-offs among management objectives. Neither the weights nor the CU scale are inferred monetary costs or empirically elicited airline utility. Native component definitions, proxy status, and scaling rules are documented in Section 4 and the accompanying technical material.

The same consequence mapping supports the two decision layers in different ways: Stage I compares baseline exposure across flight chains, whereas Stage II evaluates changes in the consequences of one chain after a modeled recovery adjustment and accounts for recovery effort. The two layers therefore share an evaluation framework without repeating the same optimization problem. Re-evaluating the seven components after an adjustment does not imply that every component responds to that adjustment.

3.3.1. Fixed-budget recovery screening

For actionable stage $g \in \mathcal{G}_A$, let $\mathcal{N}_g$ denote the candidate set under a specified screening evaluation and K_g its review budget. The consequence-based priority measures the expected baseline exposure before any *additional* local recovery target specified by this framework:

$$P^{C}_{i,g}(\boldsymbol{\omega}; P_R) = \mathbb{E}_{s \sim P_R}\left[\Phi_{\boldsymbol{\omega}}\big(\mathbf{C}_{i,g,s}(0)\big)\right].$$

The associated screening problem is

$$\begin{aligned} \max_{\{z_{i,g}\}} \quad & \sum_{i \in \mathcal{N}_g} P^{C}_{i,g} z_{i,g} \\ \text{s.t.} \quad & \sum_{i \in \mathcal{N}_g} z_{i,g} \le K_g, \qquad z_{i,g} \in \{0,1\}. \end{aligned} \tag{3}$$

With nonnegative priorities and one review slot per chain, the solution selects the K_g highest-scoring candidates, using a predetermined tie-break. Replacing $P^{C}_{i,g}$ by $P^{D}_{i,g}$ changes the priority signal while retaining the same candidates, review budget, and reference consequence evaluator. The objective is to cover high-consequence disruptions for subsequent diagnosis; it does not require Stage I to have already identified the largest achievable recovery benefit for each chain. Using a fully optimized recoverable-value ranking at screening would instead assume that much of Stage-II assessment had already been undertaken. The primary quarterly evaluation remains a retrospective, stage-specific benchmark unless concurrent competition and actual capacity are independently established. PRE and TURN are not pooled into a single real-time queue. Whether a high-consequence shortlist also captures recoverable opportunities is evaluated separately.

3.3.2. Local-recovery targeting

For a shortlisted chain with valid local support, let $u \in \mathcal{U}_{i,g}$ denote a candidate effective recovery target, with $0 \in \mathcal{U}_{i,g}$. The nominal target u is distinct from the advance actually achievable in a particular state: scenario s supplies nonnegative modeled recovery headroom $H_{i,g,s}$, so that

$$r_{i,g,s}(u) = \min\{u, H_{i,g,s}\}, \qquad AOBT_{i,g,s}(u) = AOBT_{i,g,s}(0) - r_{i,g,s}(u).$$

The first expression distinguishes the selected target from the scenario-specific realized improvement, while the second updates the successor's actual off-block time. All seven consequences are recomputed using their native response functions under the adjusted state, but some components may remain invariant. For example, if additional taxi-out delay is unchanged by the off-block adjustment, the Resource component cannot be said to improve directly. Nevertheless, changing its weight can alter the relative importance of responsive consequences and effort, thereby affecting the chosen target. For representation R with conditional state distribution P_R, the Stage-II objective is

$$J_{i,g,\boldsymbol{\omega}}(u; P_R) = \mathbb{E}_{s\sim P_R}\left[\Phi_{\boldsymbol{\omega}}\left(\mathbf{C}_{i,g,s}(u)\right)\right] + c_i^A(u), \tag{4}$$

where the first term is the expected multidimensional consequence after the target is applied and $c_i^A(u)$ is a normative effort penalty for selecting that target. This term is not directly measured expenditure. The optimal target and its model-relative recoverable value are

$$\begin{aligned} u^*_{i,g,\boldsymbol{\omega}}(P_R) &= \arg\min_{u\in\mathcal{U}_{i,g}} J_{i,g,\boldsymbol{\omega}}(u; P_R), \\ V_{i,g,\boldsymbol{\omega}}(P_R) &= J_{i,g,\boldsymbol{\omega}}(0; P_R) - J_{i,g,\boldsymbol{\omega}}\left(u^*_{i,g,\boldsymbol{\omega}}(P_R); P_R\right). \end{aligned} \tag{5}$$

Ties are resolved in favor of the smaller target. As zero is always feasible, the model-optimal recoverable value is nonnegative. Yet u^* is an effective local-recovery target rather than a validated resource-dispatch instruction, and V is a modeled objective improvement rather than an observed economic benefit. The zero-target case excludes only *additional* adjustments represented in this framework; it does not presume the absence of historical airline interventions. Likewise, the headroom restriction covers the modeled timing limits, not all crew, gate, maintenance, and airport-capacity requirements. Detailed effort parameters, physical consistency checks, and component-wise response audits are addressed in Section 4 and the supplementary material.

3.4. Decision Sufficiency and Incremental Decision Value

Different information representations may change estimated operating states and expected consequences without changing the resulting recovery choice. We therefore distinguish exact decision equivalence from reference-objective loss: the former asks whether the same option is selected, while the latter assesses the consequences of any difference using a common evaluator. The loss is model-relative, not the classical ex ante value of acquiring new observations and not the causal effect of an unimplemented recovery action.

3.4.1. Decision equivalence and reference-objective loss

Let R_r and R_a be reference and alternative representations. Under the same objective weights and admissible decision sets, Stage-I outputs are the shortlists $\mathcal{H}_g(R)$ and Stage-II outputs are the selected targets $u^*_{i,g}(R, \boldsymbol{\omega})$. Exact decision equivalence is

$$\begin{aligned} R_a \sim_g^I R_r &\iff \mathcal{H}_g(R_a) = \mathcal{H}_g(R_r), \\ R_a \sim_{i,g}^{II} R_r &\iff u^*_{i,g}(R_a, \boldsymbol{\omega}) = u^*_{i,g}(R_r, \boldsymbol{\omega}). \end{aligned} \tag{6}$$

Decision equivalence does not require identical predicted distributions or scores. Conversely, different selections may be tied or nearly tied under the reference criterion. Selection changes and their objective consequences are therefore distinct outcomes and must be reported separately.

Let $\mathcal{H}_g^r$ and $\mathcal{H}_g^a$ be the Stage-I shortlists, with retained values

$$A_g^r = \sum_{i\in\mathcal{H}_g^r} P_{i,g}^{C,r}, \qquad A_g^a = \sum_{i\in\mathcal{H}_g^a} P_{i,g}^{C,r}.$$

For $A_g^r > 0$, the normalized screening loss is

$$L_g^{\text{att}} = 1 - \frac{A_g^a}{A_g^r}.$$

Both shortlists use the same candidate population, budget, objective weights, and reference consequence priority for evaluation. The loss is nonnegative when the reference shortlist maximizes that criterion over the common population and review capacity. Replacement counts describe how much the shortlist changes; L_g^{att} describes the amount of reference consequence coverage lost. Neither is a direct measure of realized airline savings.

For Stage II, let $u_{i,g}^r$ and $u_{i,g}^a$ be the two targets and evaluate both using the *reference* objective $J_{i,g,\boldsymbol{\omega}}^r$:

$$\Delta_{i,g,\boldsymbol{\omega}}^{\text{rec}} = J_{i,g,\boldsymbol{\omega}}^r(u_{i,g}^a) - J_{i,g,\boldsymbol{\omega}}^r(u_{i,g}^r).$$

For a paired, actionable cohort C, define

$$L^{\text{rec}}(C, \boldsymbol{\omega}) = \frac{\sum_{(i,g)\in C} \Delta_{i,g,\boldsymbol{\omega}}^{\text{rec}}}{\sum_{(i,g)\in C} V_{i,g,\boldsymbol{\omega}}^r}, \tag{7}$$

only for a positive reference-value denominator and a paired cohort with common support and feasibility. The ratio is not intrinsically bounded above by one, and a zero denominator produces an undefined ratio rather than an observed zero loss. Absolute CU losses, denominator values, and supported sample sizes must accompany normalized results. Each $\boldsymbol{\omega}$ has its *own* reference optimum and reference-value denominator; a Balanced solution cannot serve as the optimum for a different preference criterion. Zero loss under the reference representation follows by construction from its model-defined optimality and cannot establish that it is the best representation in real-world airline recovery. Stage-I and Stage-II losses also have different objects and denominators: their magnitudes should not be directly compared as if they measured a single realized benefit.

3.4.2. Decision boundaries and representation perturbations

The relevant comparison is across the active decision boundary. For a same-scale Stage-I comparison, write $\widetilde{P}_{i,g} = P_{i,g} + \delta_{i,g}^I$. Define the reference cutoff margin and the maximum relative perturbation across selected and unselected cases by

$$m_g = \min_{i\in\mathcal{H}_g^r,\, j\notin\mathcal{H}_g^r} (P_{i,g} - P_{j,g}),$$
$$\Delta_g^I = \max_{i\in\mathcal{H}_g^r,\, j\notin\mathcal{H}_g^r} \left|\delta_{i,g}^I - \delta_{j,g}^I\right|.$$

A sufficient condition for an unchanged shortlist is

$$\Delta_g^I < m_g \quad \Longrightarrow \quad \mathcal{H}_g^a = \mathcal{H}_g^r. \tag{8}$$

For Stage II, let $u_{i,g}^r$ be the reference optimum, $\gamma_{i,g}$ the smallest reference-objective gap to another feasible target, and $\delta_{i,g}^{II}(u)$ the alternative-minus-reference objective perturbation. Define

$$\gamma_{i,g} = \min_{u\in\mathcal{U}_{i,g},\, u\neq u_{i,g}^r} \left[J_{i,g}^r(u) - J_{i,g}^r(u_{i,g}^r)\right],$$
$$\Delta_{i,g}^{II} = \max_{u\in\mathcal{U}_{i,g},\, u\neq u_{i,g}^r} \left|\delta_{i,g}^{II}(u) - \delta_{i,g}^{II}(u_{i,g}^r)\right|.$$

Then

$$\Delta_{i,g}^{II} < \gamma_{i,g} \quad \Longrightarrow \quad u_{i,g}^a = u_{i,g}^r. \tag{9}$$

These are standard *sufficient*, not necessary, stability conditions; failure of a certificate does not imply that a choice changes. Tie cases and singleton feasible sets follow the common decision rules and are documented with the derivations in the Supplementary Material. Both conditions concern comparable, same-scale objective perturbations. Raw delay minutes and normalized consequence scores are not on a common scale, so the Delay–Consequence comparison must be evaluated through selected shortlists and retained reference value rather than direct score subtraction.

3.4.3. Decision objectives and distributional dependence

Let P_J and P_M be Joint and valid Marginal couplings with identical weighted primitive marginals, the same target set, and the same effort specification. Define the objective difference for a fixed target u:

$$\begin{aligned} e_{\boldsymbol{\omega}}(u) &= J_{\boldsymbol{\omega}}(u; P_J) - J_{\boldsymbol{\omega}}(u; P_M) \\ &= \sum_{d \in \{F,P,R\}} \omega_d \left[\mathbb{E}_{P_J} D_d(u, S) - \mathbb{E}_{P_M} D_d(u, S) \right]. \end{aligned} \tag{10}$$

An individual objective-level difference need not alter the preferred target. Let u_J^* minimize $J_{\boldsymbol{\omega}}(\cdot; P_J)$ and, for another candidate v, define $g_{J,\boldsymbol{\omega}}(v) = J_{\boldsymbol{\omega}}(v; P_J) - J_{\boldsymbol{\omega}}(u_J^*; P_J)$. Algebraically,

$$\begin{aligned} &J_{\boldsymbol{\omega}}(v; P_M) - J_{\boldsymbol{\omega}}(u_J^*; P_M) \\ &\quad = g_{J,\boldsymbol{\omega}}(v) - \left[e_{\boldsymbol{\omega}}(v) - e_{\boldsymbol{\omega}}(u_J^*) \right]. \end{aligned} \tag{11}$$

Thus, if

$$e_{\boldsymbol{\omega}}(v) - e_{\boldsymbol{\omega}}(u_J^*) > g_{J,\boldsymbol{\omega}}(v),$$

Marginal strictly prefers v to the original Joint-optimal target u_J^*; equality is governed by the fixed tie-break. The condition does not assert that v must be Marginal's global optimum. Dependence can alter the absolute evaluation of every candidate yet leave the preferred target unchanged if those evaluations shift by the same constant. What matters for selection is the *difference in dependence perturbations across feasible targets*, compared with the original objective gap. Domain weights determine the relative importance of consequences, while cross-state interactions determine which dependence patterns affect those consequences. Changing weights alone does not guarantee any new dependence sensitivity: a domain invariant to u may affect the balance between responsive domains and effort without providing a direct improvement in that domain. These relationships provide conditional mechanisms for objective-by-distribution comparisons rather than novel general optimization theorems or predetermined empirical findings.

3.4.4. Conditions for marginal sufficiency

A limiting benchmark establishes one condition under which marginal information is sufficient. Suppose that, for each fixed feasible target u, the complete consequence criterion is additive across unresolved primitive states; the Marginal and Joint representations have identical weighted one-dimensional marginals; and both use the same distribution-independent feasible set and recovery-effort term. Then

$$J_{\boldsymbol{\omega}}(u; P_J) = J_{\boldsymbol{\omega}}(u; P_M), \qquad \forall u \in \mathcal{U}. \tag{12}$$

The expected evaluation of each fixed feasible target is therefore identical, and the selected optimum also coincides under a common tie-break. This is a sufficient condition for marginal sufficiency, not a necessary condition: even a nonadditive consequence may yield identical decisions when its relative perturbation cancels across candidates. Airline recovery may depart from separability because off-block and taxi-out delays jointly enter threshold-based passenger or service exposures, while scenario-dependent headroom couples predecessor in-block timing with successor off-block timing. These mechanisms make additional dependence value *possible*, but they do not guarantee a different optimum. Their effects must be established through valid empirical comparisons rather than asserted from the existence of coupling alone.

As operations roll forward, both the evidence available to the decision maker and the remaining headroom or feasible recovery set may change. Decisions at different epochs therefore do not necessarily concern the same optimization problem. Their differences should be analyzed separately from fixed-epoch representation contrasts and must not, by themselves, be interpreted as causal gains from information revelation.

4. Empirical Implementation

This section describes the empirical implementation of the rolling airline recovery framework. Using U.S. airline operating data for 2017–2022, we construct aircraft-linked flight chains, reconstruct information available at successive decision times, and evaluate recovery screening and local-recovery targeting against a common multidimensional consequence criterion. The six annual evaluations cover pre-pandemic operations, the COVID-19 disruption, and the subsequent recovery of airline activity, allowing the decision effects of operating information to be examined across distinct operating environments. The original 2019 evaluation remains an identifiable benchmark within this broader design. We describe the data and temporal protocol, operating-state reconstruction, consequence measures, decision implementation, and controlled comparisons. For brevity, (i, g) denotes a flight-chain–stage case at its prespecified canonical decision epoch; other admissible epochs within the chain are reserved for the rolling analysis.

4.1. Operational Data and Temporal Evaluation Design

4.1.1. Decision-time data and aircraft-linked episodes

The primary operating unit is a predecessor–successor flight chain operated by the same aircraft. U.S. Bureau of Transportation Statistics (BTS) On-Time Performance records supply scheduled and realized flight milestones, aircraft continuity identifiers, and taxi-related observations. Airport-related weather information is drawn from the National Oceanic and Atmospheric Administration (NOAA) Integrated Surface Database. BTS DB1B and T-100 records provide historical references for passenger markets, itinerary connections, segment traffic, and capacity, which support the construction of exposure-based consequences.

These sources have distinct roles. Flight operating records describe the state of a particular aircraft-linked episode, whereas historical passenger and route references characterize the exposure associated with that episode. The latter are not live passenger itineraries, observed resource assignments, or realized recovery costs. Feature availability is established at each decision epoch: an operating milestone enters the information set only after it has occurred and become available, unresolved milestones are represented probabilistically, and historical statistical references are subject to their applicable publication and freezing rules. Realized outcomes are also retained for ex post validation but are not substituted for information unavailable when a decision is evaluated.

Eligible predecessor–successor episodes are replayed on a five-minute grid. Each node records admissible evidence and the operating stage—PRE, TURN, TAXI, or COMP—determined by the realization of predecessor in-block, successor off-block, and subsequent milestones. Prespecified canonical-epoch rules extract comparable flight-chain–stage cases from this rolling sequence. The recovery screening and local-target decisions are evaluated principally at PRE and TURN, while later stages contribute to state reconstruction and supplementary analyses. Canonical-epoch selection is distinguished from the descriptive tracking of all valid epochs within a chain.

The resulting data hierarchy separates source flights, aircraft-linked episodes, rolling nodes, canonical chain–stage cases, Stage-I screening candidates, and Stage-II supported cases. Multiple rolling nodes from one chain are not independent operating episodes, and screening eligibility does not automatically imply valid local-recovery support. Counts, decision-time availability, and reasons for exclusion are retained at each level. Detailed aircraft linking, field mappings, temporal admissibility, and episode construction are documented in the Supplementary Material.

4.1.2. Chronological evaluation design and multi-year coverage

State estimation and held-out decision evaluation are conducted independently for each of the six operating years, 2017–2022. Each year follows the same chronological partition: January–June for model estimation and historical-reference construction, July for probability calibration, August–September for model development and specification of evaluation rules, and October–December for held-out testing. Rolling observations from the same aircraft-linked chain remain within one data partition, and decision features respect their corresponding information cutoffs. The previously established 2019 evaluation and its frozen outputs retain a traceable relationship to the six-year protocol.

The study window spans three operating periods. The years 2017–2019 form a pre-pandemic benchmark in which to assess repeatability across years of broadly regular airline operations. The disruption associated with COVID-19 in 2020 provides an operating environment marked by pronounced changes in air transport demand and network activity. The years 2021–2022 represent a period of recovering traffic and continued operational adjustment. These periods enable examination of whether the decision relevance of richer information varies with operating circumstances; year-to-year contrasts describe empirical differences rather than the isolated causal effect of an external shock.

Aircraft-linking rules, canonical decision epochs, primitive-state definitions, consequence components, and the two recovery decision tasks follow a common protocol across years. Predictive parameters and probability calibrations

are estimated separately from each year's admissible training and calibration observations. The resulting evaluations address how a shared recovery-decision framework performs under different annual operating conditions, not the cross-year transfer performance of a predictor fitted once and then deployed unchanged. E1–E3 are implemented within each year where the requisite decision support is available; E4 examines the resulting between-year and within-chain heterogeneity.

Cross-year consequence comparisons use two complementary measurement tracks. The *year-local* track normalizes consequences using each year's training references and supports within-year comparisons under that year's evaluation scale. A separately frozen *fixed-common* track applies a shared reference scale across all six years to support comparisons of absolute consequence evaluations in common units. Eligible populations and local-recovery support are established separately for each year and operating stage. Consequence scales and the associated recovery-effort terms remain internally consistent within each measurement track, as described in Section 4.3.

4.2. Reconstructing Decision-Time Operating States

4.2.1. Stage-conditioned state reconstruction

As operating milestones become realized, the conditional state is reconstructed from the information available at the decision epoch. Only unresolved primitives are predicted: predecessor in-block timing is modeled using a discrete-hazard representation, while successor off-block delay and additional taxi-out delay use hurdle-type distributions that accommodate both zero and positive delay outcomes. The primitive states are conditioned in their operating sequence so that new observations reduce the remaining uncertainty without resampling quantities already realized.

At PRE, all three primitive states may remain unresolved. At TURN, predecessor in-block timing is factual and enters the subsequent conditional evaluation as observed evidence. At TAXI, successor off-block timing is also realized, leaving the remaining taxi-out uncertainty to be represented. This stage-conditioned treatment preserves the operating information that would have been available at the corresponding decision time.

For supported conditional states, the original Joint interface consists of 64 calibrated weighted scenarios, each with a coherent set of primitive state values and a nonnegative probability weight; weights sum to one. The model-specific compound successor take-off delay is formed from successor off-block delay and additional taxi-out delay. This constructed quantity follows the state convention in Section 3.2, rather than being identified with every administrative definition of observed departure delay.

Held-out prediction accuracy, probabilistic calibration, and state-support checks provide evidence on the reliability of these conditional reconstructions. Nodes with unavailable required state information or unsupported probability distributions are recorded explicitly rather than assigned zero consequences. The conditional distributions, sampling and calibration procedures, temporal origins, and support tests are given in the Supplementary Material. State definitions and validation rules remain common across years, while model parameters and calibrations are fitted independently within each annual partition.

4.2.2. Current and History representations

Two temporal-information representations are constructed at the same decision epoch. *Current* uses the contemporaneously admissible operating state and frozen historical references. *History* additionally encodes the legally available trajectory of preceding operating observations. The established History model uses a single-layer unidirectional gated recurrent unit (GRU) with hidden dimension 16. Both representations predict the same unresolved primitives and feed the same consequence and decision interfaces; neither may access subsequent realized milestones.

The original Current+Joint (CJ) versus History+Joint (HJ) comparison measures the decision effects of these implemented predictor specifications. Because adding a history encoder can also change model architecture and parameter capacity, this difference is not identified solely as the effect of historical information. A matched-capacity control is therefore included in the extended evaluation design, comparing access to the admissible trajectory while holding predictive targets, output distributions, temporal partitions, training procedures, and model complexity as comparable as practicable. Its purpose is to strengthen attribution to temporal representation, not to claim that all differences in model inductive bias have been eliminated.

The six-year evaluation retains common architecture-selection and interface rules while estimating prediction and calibration parameters within each year's legitimate data partitions. The full trajectory construction, GRU specifications, and matched-capacity diagnostics are provided in the Supplementary Material.

Table 1
Native consequence components and their empirical interpretation.

Component	Native construction	Unit	Operating interpretation
$F_{\text{continuity}}$	Predecessor in-block and turnaround-continuity proxy; temporal formula pending reconciliation	min (to verify)	Aircraft turnaround and continuity pressure
$F_{\text{execution}}$	D^{OB}	min	Successor off-block execution delay
$F_{\text{propagation}}$	$D^{TO}E^{down}$	exposure-min	Downstream operating-delay exposure
P_{time}	$N^{pax}D^{TO}$	passenger-min	Passenger-weighted time exposure
$P_{\text{itinerary}}$	$N^{pax}s^{conn}\mathbf{1}\{D^{TO} > 45\}$	event exposure	Connection-sensitive itinerary exposure
P_{service}	$N^{pax}\mathbf{1}\{D^{TO} \geq 180\}$	event exposure	Prolonged-disruption service exposure
$R_{\text{operating}}$	D^{TX}	min	Additional taxi-out operating burden

4.2.3. Point, Marginal, and Joint representations

Three uncertainty representations are constructed from the same admissible conditional state. *Point* selects an internally coherent representative scenario using the frozen weighted joint-medoid rule. *Joint* retains the original weighted joint scenarios and their cross-state dependence. *Marginal* preserves the weighted marginal distribution of each unresolved primitive while modifying the original coupling among primitives. These alternatives differ in retained probabilistic structure, rather than representing successive arrivals of new operating observations.

The Marginal–Joint comparison requires equality of the weighted one-dimensional probability distributions, not merely equality of possible values or approximate equality of moments. Each marginal reconstruction is therefore subject to probability-mass, normalization, and numerical-accuracy checks. In particular, permuting coordinates against unchanged unequal scenario weights does not in general preserve the weighted marginals. Although the original Joint interface uses 64 scenarios, a valid Marginal coupling need not retain the same scenario indices or discrete carrier.

Statistical validity and operational feasibility are examined separately. Modifying dependence may produce state combinations inconsistent with the required operating order or other support constraints. Such constructions are interpreted as diagnostic probability couplings where appropriate, not as necessarily deployable operating-state forecasts. All representations retain realized milestones and must enter subsequent paired comparisons on a common, valid decision-support population. The coupling algorithm, weighted-marginal diagnostics, operating-support checks, and any approximation errors are documented in the Supplementary Material. A dependence-specific interpretation is conditional on these checks.

4.3. Consequence Construction

4.3.1. Operational consequence components

Conditional operating states are converted into seven consequence components grouped under Flight, Passenger, and Resource. Each component combines decision-time admissible operating states, legally available or frozen historical exposure references, and prespecified consequence rules. Operating states describe realized or unresolved milestones within the flight chain; exposure references distinguish the passenger and downstream operating implications of otherwise similar delays. The seven measures are model-defined proxies for operational consequences, rather than independently observed accounts of monetary losses, passenger misconnections, or recovery expenditures. Table 1 summarizes their native constructions and interpretation.

Flight consequences represent aircraft turnaround continuity, successor execution delay, and downstream delay propagation. Passenger consequences comprise time exposure, connection-sensitive itinerary exposure, and service exposure associated with prolonged disruption. The Resource measure captures additional taxi-out delay as a local operating burden. The downstream factor E^{down}, historical passenger reference N^{pax}, and connection-exposure proportion s^{conn} are distinguished from realized passenger-level or downstream disruption outcomes. The itinerary indicator uses a strict threshold above 45 minutes, while the prolonged-disruption indicator includes delays of 180 minutes or more. These event cutoffs are prespecified measurement conventions.

The continuity measure requires confirmation of the temporal origin shared by predecessor in-block timing and the turnaround reference. In particular, subtracting a duration from an absolute timestamp would not yield a coherent turnaround-pressure quantity without an appropriate reference transformation. Its native formula and any affected results therefore remain subject to scientific reconciliation. The other consequence rules likewise require componentwise provenance and support checks; the table records their model definitions rather than independent evidence of incurred losses.

A component's role in baseline screening also differs from its response to a local recovery target. Certain consequences vary continuously with a modeled reduction in successor off-block delay, threshold components respond only when the relevant event boundary is crossed, and components unaffected by the adjusted state may remain constant. Thus, substantial baseline exposure in a domain does not imply that local recovery can directly reduce all consequences in that domain. The implementation and componentwise action-response audit are described in Section 4.4 and the Supplementary Material.

4.3.2. Reference consequence criterion for recovery decisions

Because the native components have different measurement units, they are standardized before aggregation into a common dimensionless consequence scale. For continuous components with valid positive support, the normalization reference is the median of positive observations in the permissible training period:

$$\xi_k = \mathrm{Median}_{\mathrm{Train}}(q_k \mid q_k > 0), \qquad Z_k = q_k/\xi_k, \quad \xi_k > 0.$$

The event-exposure components use their separately specified frozen event scales. Valid zero consequences are distinguished from absent observations and unsupported reference scales. Standardized components enter the hierarchical criterion defined in Eq. (2).

The primary *Balanced* criterion assigns equal weight to the three domains, Flight, Passenger, and Resource, while the three Flight and three Passenger components receive equal shares within their respective domains. Balanced therefore denotes equal *domain* weights, not equal weights across all seven components. Normalization determines the measurement scales, while domain weights specify the normative trade-offs between operating objectives. The resulting consequence unit (CU) is dimensionless; neither CU nor the criterion weights should be interpreted as observed monetary loss or estimated airline utility. Alternative Flight-, Passenger-, and Resource-emphasis criteria are evaluated as changes to the management objective, without changing operating evidence.

For the multi-year analyses, *year-local* normalization uses training-period reference scales estimated within the respective year. The supplementary *fixed-common* track uses one reference vector, frozen before assessing held-out outcomes, across all six years. Because scaling alters the effective contributions of consequence components to the decision objective, the consequence scales and Stage-II effort cost are specified as internally consistent evaluation systems in each track. Year-local comparisons characterize decision sufficiency under each year's own criterion; fixed-common comparisons provide a basis for interpreting absolute CU differences across years. Detailed scales, their provenance, and the associated cost conventions are given in the Supplementary Material.

4.4. Decision Implementation

4.4.1. Recovery screening

At Stage I, separate PRE and TURN candidate populations are ranked by expected compound successor take-off delay or by the baseline multidimensional consequence priority. Each information contrast is evaluated on a common candidate population using the same fixed review capacity and deterministic tie-breaking rule. Resulting shortlists are assessed under a common reference consequence criterion, so observed shortlist differences reflect priority signals rather than changes in candidate scope, review capacity, or outcome evaluation.

The nominal detailed-review capacity is set to 10% of valid candidates in each operating stage, rounded up:

$$K_g(q) = \lceil qN_g \rceil, \qquad q_0 = 0.10.$$

Candidate populations and shortlist capacities are calculated separately for each year and stage; hence the number of selected cases varies with annual eligibility. As an established benchmark, the October–December 2019 evaluation contains 29 PRE and 127 TURN candidates, yielding shortlists of 3 and 13 cases, respectively. The review fraction represents an analytical allocation of limited diagnostic attention, rather than a directly estimated AOCC staffing level or a verified contemporaneous queue capacity.

Screening comparisons report both selected cases and retained reference consequences. This Stage-I optimum is defined by baseline consequence exposure under the specified candidate set and budget, rather than by the maximum improvement subsequently available from local intervention. Whether different screening rules capture recoverable opportunities is examined separately in Section 4.5.

4.4.2. Effective local-recovery targeting

For shortlisted flight-chain–stage cases with valid local-recovery support, Stage II selects from the discrete nominal target set $\mathcal{U} = \{0, 5, 10, \dots, 45\}$ minutes. An empirical turnaround reference of $T_{\text{turn,lb}} = 41$ minutes is derived from the lower 20th percentile of historical realized turnaround times. Together with scheduled successor off-block time and scenario-specific predecessor in-block timing, it defines the earliest modeled successor off-block time and the available local recovery headroom:

$$\begin{aligned} LB^{OB}_{i,g,s} &= \max\{SOBT^{+}_{i}, T^{IB}_{i,g,s} + T_{\text{turn,lb}}\}, \\ H_{i,g,s} &= [AOBT_{i,g,s}(0) - LB^{OB}_{i,g,s}]_{+}, \\ r_{i,g,s}(u) &= \min\{u, H_{i,g,s}\}. \end{aligned}$$

The empirical turnaround bound constrains modeled timing adjustments; it is not a universal aircraft, airport, crew, or safety minimum. In addition, zero headroom produced by positive-part truncation does not by itself establish the feasibility of the underlying baseline state. Inconsistent timing or unsupported scenarios are subject to separate diagnostics.

For each candidate target, the model computes the effective reduction in successor off-block time for each conditional operating scenario, updates the corresponding state, and re-evaluates the same seven consequences. The response depends on which states the target changes: delay-related components may improve, event-exposure measures may change when thresholds are crossed, and unaffected components may remain constant. A nominal recovery-effort penalty is then added, with $\lambda_0 = 0.25$ and $U_{\max} = 45$ minutes under the reference specification. The penalty is proportional to the *selected* target rather than the realized scenario-by-scenario advance, providing a model-defined trade-off between expected consequences and additional effort.

Each information representation selects the target minimizing its own expected consequence-plus-effort objective on the common feasible target grid; ties are resolved in favor of the smaller target. Selecting $u = 0$ means no additional modeled local recovery, not that historical operations involved no other interventions. All target evaluations require valid operating-state and temporal support. Implementation diagnostics and detailed effort-cost sensitivity are provided in the Supplementary Material.

The Stage-II reference cohort is constructed separately for each year from the HJ consequence-based shortlist, subject to valid local-recovery support. Its size therefore follows the year-specific candidate population, review budget, and eligibility conditions. The original 2019 reference contains 16 shortlisted cases (3 PRE and 13 TURN) and remains separately identifiable for comparison with the frozen results. Beyond these annual reference cohorts, the full local-recovery-eligible population supports analysis of recoverable opportunities outside the selected shortlist. Cohort definitions and their distinct evaluation purposes are specified in Section 4.5.

4.5. Controlled Evaluation of Information Value

4.5.1. Information comparisons

The evaluation addresses four related empirical questions. E1 examines whether delay is sufficient for consequence-based screening; E2 assesses how temporal and probabilistic information affects screening and local recovery; E3 tests when management objectives and cross-state dependence alter the value of joint distributions; and E4 characterizes operating-stage, annual, and within-chain heterogeneity in these relationships. E1–E3 are the main decision comparisons and are evaluated independently for each supported year in 2017–2022. E4 synthesizes their variation across operating contexts rather than treating the additional years as a separate supplementary test. All comparisons identify the changing information or decision condition and maintain an appropriate common population, feasible choice set, and reference evaluator. Table 2 summarizes their management questions and outcome measures.

E1 retains the original expected-Delay versus Consequence screening benchmark and adds a Delay+Exposure comparator using admissible historical passenger, route, or downstream exposure references. The intermediate score and any fitted coefficients are specified using the permitted development data, rather than chosen using held-out recovery outcomes. All three priorities face the same candidates and review budgets. This contrast assesses how much of the additional discrimination afforded by multidimensional consequences can be explained by a simpler combination of expected delay and exposure information.

E2 takes HJ as the principal reference and compares it with CJ, History+Point (HP), and History+Marginal (HM). The paired evaluation measures Stage-I shortlist and consequence-coverage differences together with Stage-II activation, target agreement, absolute regret, and normalized reference loss. It also examines whether ranking chains

Table 2
Controlled information comparisons and the corresponding management questions.

Module	Primary contrast	Management decision question	Main outcomes
E1	Delay / Delay+Exposure / Consequence	How much consequence-based screening is retained by a simpler delay signal?	Shortlist changes, retained consequences, screening loss
E2	CJ–HJ; HP–HJ; HM–HJ; recoverability	Does information representation change review priorities or local recovery choices?	Choice agreement, reference loss, recovery-opportunity coverage
E3	Four objectives × Marginal/Joint; structural ablations	Under which objectives and state couplings does joint dependence matter?	Relative objective perturbation, selected targets, reference regret
E4	2017–2022; PRE/TURN; rolling epochs	How do these decision relationships vary across operating conditions?	Annual replication, operating heterogeneity, within-chain trajectories

by baseline consequences captures chains with large model-recoverable opportunities. The matched-capacity temporal control and validated marginal-preserving coupling are essential to the corresponding attribution analyses; evidence derived from their unverified implementations is not presented as a pure information-component effect.

E3 evaluates the Marginal and Joint recovery decisions under four prespecified domain-weight vectors: Balanced, Flight emphasis, Passenger emphasis, and Resource emphasis. For each objective, both decisions are recomputed and assessed against that objective's Joint reference. To distinguish changes in fixed-target evaluation from changes in the selected target, analyses record relative dependence perturbations, choice agreement, and reference regret. The main mechanism comparison holds an objective-independent common-supported cohort fixed; shortlist changes induced by altering the objective are evaluated separately. Thus, changing a management preference does not simultaneously alter the main sample used to assess dependence sensitivity.

E4 examines how the E1–E3 decision relationships vary across the independent annual evaluations, PRE and TURN stages, recovery headroom, threshold-adjacent states, and successive admissible epochs within a flight chain. Annual contrasts are interpreted in the context of the pre-pandemic benchmark (2017–2019), the pandemic disruption (2020), and operating recovery (2021–2022), as introduced in Section 4.1.2. Alongside the magnitude and direction of each decision effect, differences in support and in the proximity of candidates to decision boundaries inform the assessment of operating heterogeneity.

4.5.2. Held-out evaluation and inference

All comparisons use chronologically held-out cases under matching decision epochs, operating stages, and admissible information. For each year and stage, an HJ consequence-based reference shortlist is formed under the shared budget rule, and Stage-II comparisons use the cases with valid local-recovery support. The original 2019 cohort $\mathcal{C}_0$ contains 16 HJ-shortlisted cases (3 PRE and 13 TURN) and remains a separately traceable benchmark, rather than defining the sample size or membership of the six-year evaluation.

Four populations serve distinct inferential purposes. $\mathcal{C}_0$ preserves the original 2019 paired results; $\mathcal{C}_1$ is the full population with valid local-recovery support, defined independently of screening selection; $\mathcal{C}_2$ is the union of alternative screening shortlists within a fixed year, stage, review budget, and objective; and $\mathcal{C}_3$ comprises multiple admissible rolling epochs for the same chain. Annual reference shortlists provide the corresponding year-local pairing for the six-year tests. Because these populations answer different questions, valid counts, support restrictions, exclusions, and reference denominators are reported separately. In particular, the Stage-I screening population does not automatically coincide with the entire Stage-II eligible population.

The primary outcomes include shortlist agreement, recovery activation, exact target agreement, reference consequence coverage, and absolute and normalized decision loss. We also assess *recovery-opportunity coverage*: the share of reference model-recoverable value contained in a screening shortlist, relative to the full supported eligible population in the same year, stage, objective, and CU scale. For a shortlist $\mathcal{H}_K$ and full supported set $\mathcal{C}_{1,y,g}$, the measure is

$$\mathrm{Coverage@K}_{y,g,\boldsymbol{\omega}} = \frac{\sum_{i \in \mathcal{H}_K \cap \mathcal{C}_{1,y,g}} V^{HJ}_{i,g,\boldsymbol{\omega}}}{\sum_{i \in \mathcal{C}_{1,y,g}} V^{HJ}_{i,g,\boldsymbol{\omega}}},$$

when the denominator is positive. It captures modeled recovery opportunity available for subsequent review, rather than realized savings. Each paired representation contrast is evaluated on its valid common-support intersection.

The original supported-mass criterion of $m^{CS} \geq 0.90$ is retained where defined, subject to a representation-aware calculation when alternative couplings have different probability carriers. Valid zero-target and zero-loss observations are retained; undefined normalized ratios are reported separately from zero losses.

Statistical inference uses the aircraft-linked flight chain as the primary resampling unit, keeping repeated decision epochs from one chain together. The established paired analyses use $B = 2000$ percentile-bootstrap replicates. For set-valued Stage-I Top-K outcomes, bootstrap replicates instead reconstruct the candidate population, review capacity, ranking, ties, and reference consequence coverage to reflect uncertainty in the screening procedure as a whole. Airport-day or calendar-day block sensitivity can assess common operating shocks where effective cluster support is sufficient. Prediction scenarios, target alternatives, and rolling nodes are not treated as independent flight chains.

Annual estimates, eligible denominators, and uncertainty intervals are reported separately before any explicitly weighted multi-year summaries. Year-local and fixed-common CU comparisons remain within their internally consistent measurement and effort-cost systems. The original frozen outputs and newly executed controlled comparisons retain distinct result provenance, allowing changes due to scientific reconciliation to be traced.

4.5.3. Decision-boundary and sensitivity analyses

Decision-boundary analyses investigate when different representations support the same choices. Stage-I screening is evaluated at review fractions $q \in \{0.05, 0.10, 0.20, 0.30\}$, with $q_0 = 0.10$ as the main specification; shortlist replacements, retained reference consequences, and cutoff-adjacent cases are compared under a common stage-specific population. Stage II examines the gaps between candidate recovery targets and the relative objective perturbations associated with information representation, relating choice stability to the conditions in Section 3.4.

For dependence sensitivity, the four objective-weight settings are combined with an ablation of threshold consequences and state-dependent recovery headroom. The threshold mechanism concerns the 45-minute itinerary-exposure and 180-minute service-exposure events; the headroom mechanism links realized recovery to predecessor in-block and successor off-block states. Threshold off/on × Headroom coupling off/on defines four diagnostic structures. Each contrast identifies the precise altered operator and keeps the remaining consequence definitions, CU scales, effort terms, and common-supported cases aligned to the extent warranted by its design. Disabling a threshold operator is distinguished from deleting an entire consequence domain. Outcomes include fixed-target objective perturbations, changes in the selected recovery target, and regret under the appropriately specified Joint reference.

Recovery-policy sensitivity varies the effort coefficient $\lambda \in \{0.10, 0.25, 0.50, 1.00\}$, the empirical turnaround reference $T_{\text{turn,lb}} \in \{34, 41, 47\}$ minutes, and the maximum recovery target $U_{\max} \in \{25, 45, 75\}$ minutes. The extended design separates changes in target range from changes in the effort-cost slope $\lambda/U_{\max}$, which move together in the original range sensitivity. Evaluations report target selection, activation, reference recoverable value, and information-representation loss under each policy setting.

Operating heterogeneity is examined across the six annual evaluations, PRE/TURN stages, relevant weather and airport conditions, proximity to delay thresholds, and remaining recovery headroom. Within-chain analyses trace the evolution of uncertainty, recovery opportunity, selected targets, and representation loss across admissible rolling epochs. As the operating process unfolds, information availability and feasible recovery opportunity may change together; changes across epochs therefore describe their joint evolution rather than identifying an information-revelation effect in isolation. Full parameter grids, alternative support rules, component response diagnostics, and the detailed inference procedures are supplied in the Supplementary Material.

5. Empirical Results

This section examines the incremental decision value of aircraft-linked operating information beyond expected delay across the two recovery decisions. We first establish the supported six-year evaluation populations, then assess consequence-aware recovery screening and the relationship between screening priority and recoverable opportunity. We next compare temporal and probabilistic representations across decision layers, investigate the information requirements of alternative recovery objectives, and characterize variation across operating stages and annual regimes. The comparisons use chronologically held-out cases, prespecified decisions, and objective-specific reference criteria, as detailed in Section 4.5.

5.1. Evaluation Coverage and Operating Context

The evaluation covers six independently estimated and calibrated operating years, 2017–2022, with year-specific held-out tests in October–December. Table 3 distinguishes the PRE and TURN screening populations from the cohorts

Table 3
Annual held-out screening populations and local-recovery evaluation support, 2017–2022.

Year	PRE N	TURN N	PRE K	TURN K	Full Stage-II eligible	Paired Stage-II supported
2017	1 310	3 899	131	390	5 209	521
2018	1 345	3 873	135	388	5 218	523
2019	1 252	3 888	126	389	5 140	515
2020	675	4 053	68	406	4 728	474
2021	1 297	3 930	130	393	5 227	523
2022	1 422	3 884	143	389	5 306	532

N is the number of canonical stage-specific screening candidates and $K = \lceil q_0 N \rceil$ the nominal $q_0 = 0.10$ review capacity, both computed separately by stage. Full Stage-II eligibility is defined independently of shortlist membership; in these years every canonical PRE/TURN candidate satisfies the full common-support condition used for local-recovery evaluation, so this column equals the sum of the two screening populations. The paired Stage-II column is the annual History+Joint reference cohort (the HJ consequence shortlist intersected with local-recovery support); all shortlisted cases retained valid recovery support in every year, so these cohorts equal the annual $K_{y,\mathrm{PRE}} + K_{y,\mathrm{TURN}}$ selections. The original 2019 PRE/TURN shortlist of 3/13 is a provenance benchmark, not a substituted annual eligibility count.

eligible for local-recovery evaluation and from the cases satisfying paired representation-support requirements. These populations answer different decision questions; their sizes must not be used interchangeably. The original 2019 HJ-selected 16-case cohort is retained as a traceable historical benchmark rather than imposed as the Stage-II sample size for other years.

Within each year, decision comparisons use the appropriate year-local consequence scale; analyses of absolute consequences across years use the separately frozen fixed-common scale. Annual, stage-specific support and any departures from the prespecified inclusion rules are reported before interpreting differences between operating environments.

Across the six held-out windows the PRE screening populations span 675 (2020) to 1 422 (2022) candidates, while the TURN populations remain comparatively stable at 3 873–4 053 candidates. The nominal review capacities inherit that asymmetry: 68–143 PRE positions versus 388–406 TURN positions. The 2020 PRE population is the smallest of the six years, consistent with the contraction of scheduled aircraft-linked turnover in the disruption year rather than with a change in eligibility rules; TURN eligibility is governed by realized predecessor arrivals and shrinks much less. These composition differences are the operating context against which the stage-specific screening and local-recovery results below are read; they are not treated as an identified effect of the pandemic.

5.2. Delay-based Screening and Consequence Coverage

The first comparison asks how much incremental screening value is obtained when a scalar decision-time expected-delay priority is replaced by multidimensional consequence information. Overall rank association is informative but is not sufficient to establish substitutability under a fixed review budget. The operational test is whether the two priorities admit the same chains into the stage-specific top-K shortlist and, when they do not, how much reference consequence coverage is lost.

Figure 3 reports the six-year PRE/TURN screening comparison. Each year and stage is evaluated over a common candidate population and the same K, with shortlist overlap, replacements, retained reference consequences, and normalized screening loss L^{att}. Rank association is reported as a supporting diagnostic, not substituted for the screening outcome. PRE and TURN shortlists are constructed separately, even where stage-specific outcomes are summarized together.

At the nominal review budget, expected delay displaces 37–53% of PRE positions and 40–55% of TURN positions, and the displaced positions carry $L^{\mathrm{att}} = 0.101$–0.184 of the reference consequence value in PRE and 0.107–0.181 in TURN. The minimum loss occurs in 2020 (0.1006, 95% CI [0.0712, 0.1461] in PRE; 0.1068, [0.0931, 0.1216] in TURN) and the maximum in 2022 (0.1840, [0.1506, 0.2163] in PRE; 0.1811, [0.1618, 0.1996] in TURN). The intervals exclude zero in every year and stage, so the boundary disagreement between the two priorities is not a small-sample artifact: under the nominal 10% review capacity, 27–76 PRE positions and 154–215 TURN positions change in each year.

An intermediate Delay+Exposure priority adds admissible historical passenger or downstream exposure references to the expected-delay signal. Adding the passenger exposure reference reduces the screening loss by 6–45% in PRE and 20–38% in TURN, leaving $L^{\mathrm{att}} = 0.063$–0.172 (PRE) and 0.067–0.145 (TURN); a substantial residual boundary

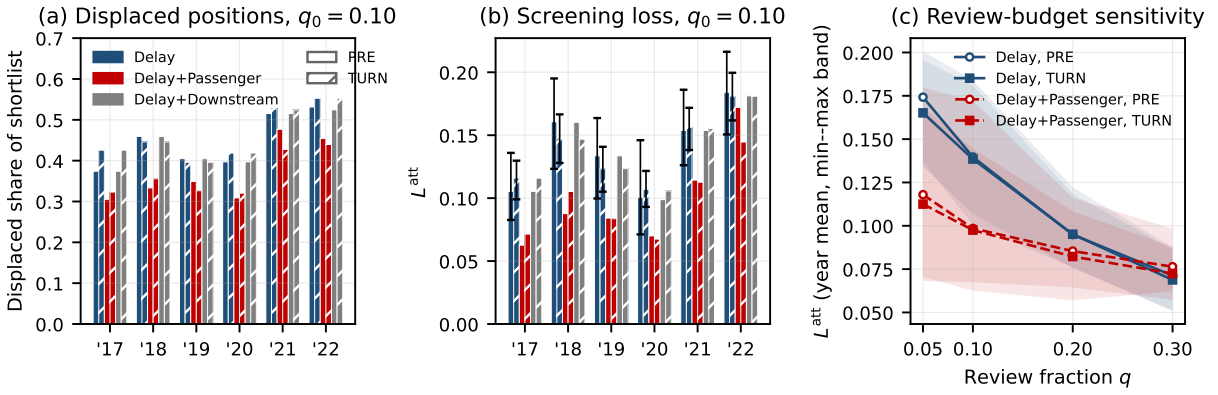


Figure 3: Expected-Delay versus Consequence screening across 2017–2022, separately for PRE (solid) and TURN (hatched). Panel (a) reports the share of reference shortlist positions displaced by each priority at the nominal review budget $q_0 = 0.10$; panel (b) reports the normalized reference-consequence loss L^{att} at the same budget, with episode-cluster percentile bootstrap intervals ($B = 2000$) for the Delay priority; panel (c) reports the review-budget sensitivity of L^{att} averaged across years, with the full annual range shown as a band.

disagreement remains in every year, and in 2022 the passenger multiplier removes only a small part of the gap (0.1840 → 0.1721 in PRE). Adding the downstream propagation reference behaves differently: the resulting priority reproduces the delay shortlist almost exactly, with $|\Delta L^{\text{att}}| \leq 0.003$ relative to expected delay in all twelve year-stage settings. The review-budget sensitivity in Fig. 3(c) shows that the disagreement is concentrated at tight review budgets: the year-mean L^{att} of the delay priority falls from 0.174 to 0.071 (PRE) and from 0.165 to 0.069 (TURN) as the review fraction increases from 0.05 to 0.30, and the Delay+Passenger and Delay priorities converge to within 0.006 at $q = 0.30$ (year-mean gap 0.0051 in PRE and 0.0037 in TURN).

5.3. Screening Priority and Recoverable Opportunity

Consequence-based screening prioritizes baseline disruption exposure, whereas local recovery evaluates the improvement still available after accounting for feasibility and the effort of pursuing an additional effective off-block target. A highly prioritized chain need not have a positive model-optimal recovery target; likewise, recoverable value is not interchangeable with baseline consequence. The relationship is therefore examined using the valid Stage-II population rather than being inferred solely from a consequence-selected shortlist.

For each supported chain-stage case in the annual reference cohort, we obtain the HJ reference target u^*, activation indicator $\mathbf{1}\{u^* > 0\}$, and model-recoverable value $V = J^{HJ}(0) - J^{HJ}(u^*)$. Figure 4 reports the resulting activation and recoverable-value distributions under the Balanced objective, together with the objective-wise activation share. Under Balanced, the reference target is positive for 82–92% of the annual reference cohorts, and the share of cases whose model-recoverable value is exactly zero is 8.4–17.7%. The median recoverable value per case ranges from 0.018 CU (2017) to 0.136 CU (2018), against annual aggregate reference values $\sum_i V_i^{HJ}$ of 40.4–85.1 CU. Activation is higher in TURN than in PRE in every year (0.83–0.93 versus 0.77–0.88), and the zero-recoverable-value share is correspondingly higher in PRE than in TURN in every year (PRE 0.115–0.229 versus TURN 0.074–0.175).

The distributions also document how close the reference decisions are to their decision boundaries: the median gap between the optimal target's objective value and the best alternative is 0.010–0.021 CU across years and objectives, which is the same order as the objective perturbations examined in Section 5.5. We therefore do not treat target stability as evidence of information sufficiency anywhere in this section; it is reported alongside the magnitude of the underlying objective gaps.

5.4. Decision Effects of Operating Information

We next examine whether the incremental decision value of operating information differs between the two decision layers. The paired contrasts use History+Joint (HJ) as the reference, with Current+Joint (CJ), History+Point (HP), and History+Marginal (HM) as alternatives. Each comparison fixes the relevant decision epoch, feasible set, objective, and reference evaluator, and is computed on the paired common-support intersection of the two representations at the

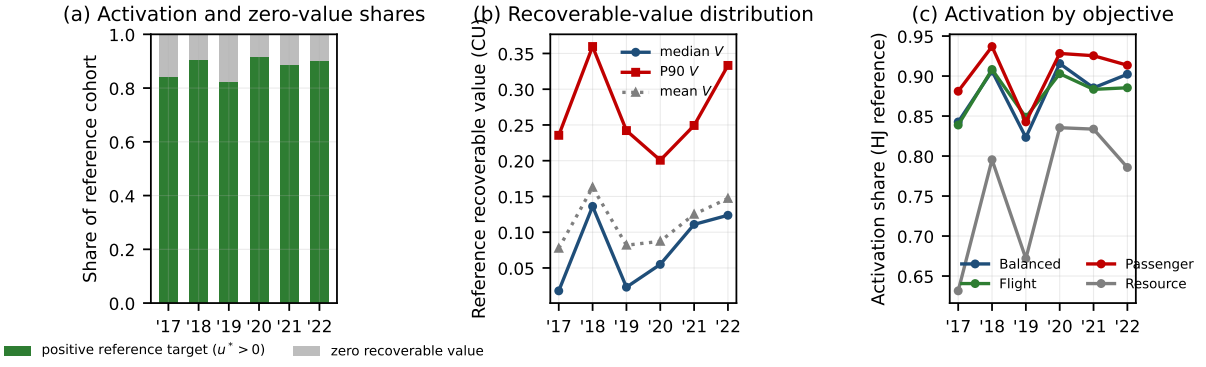


Figure 4: Baseline consequence priority and model-recoverable opportunity in the annual Stage-II reference cohorts. Panel (a) reports the share of cases with a positive History+Joint reference target and the share with zero recoverable value under the Balanced objective; panel (b) reports median, mean, and P90 per-case recoverable value V^{HJ}; panel (c) reports the activation share under the four prespecified objective-weight settings. All panels use the same annual reference cohorts as Table 3.

Table 4
Six-year decision effects of temporal and probabilistic representations.

Year	Contrast	Stage-I displaced share of shortlist	L^{att} [95% CI]	Stage-II exact agreement A_0	L^{rec} [95% CI]
2017	CJ–HJ	0.2361	0.0320 [0.0255, 0.0392]	0.342	0.6676 [0.6028, 0.7382]
	HP–HJ	0.8580	0.4986 [0.4708, 0.5241]	0.226	0.9145 [0.8780, 0.9447]
	HM–HJ	0.0192	0.0010 [0.0004, 0.0018]	0.693	0.1095 [0.0818, 0.1416]
2018	CJ–HJ	0.1683	0.0233 [0.0180, 0.0292]	0.239	0.6194 [0.5622, 0.6833]
	HP–HJ	0.8757	0.5050 [0.4783, 0.5309]	0.130	0.9814 [0.9718, 0.9892]
	HM–HJ	0.0191	0.0011 [0.0001, 0.0020]	0.585	0.0935 [0.0718, 0.1178]
2019	CJ–HJ	0.1806	0.0253 [0.0214, 0.0330]	0.353	0.6957 [0.6359, 0.7612]
	HP–HJ	0.8854	0.5291 [0.5052, 0.5543]	0.241	0.9321 [0.8975, 0.9616]
	HM–HJ	0.0194	0.0009 [0.0004, 0.0015]	0.730	0.0853 [0.0623, 0.1129]
2020	CJ–HJ	0.8987	0.5060 [0.4820, 0.5285]	0.086	0.9977 [0.9944, 1.0006]
	HP–HJ	0.8418	0.4313 [0.4071, 0.4563]	0.171	0.8963 [0.8721, 0.9192]
	HM–HJ	0.0127	0.0006 [0.0000, 0.0012]	0.631	0.0906 [0.0727, 0.1131]
2021	CJ–HJ	0.1224	0.0101 [0.0075, 0.0134]	0.495	0.3102 [0.2616, 0.3662]
	HP–HJ	0.8509	0.4134 [0.3865, 0.4367]	0.153	0.9535 [0.9365, 0.9680]
	HM–HJ	0.0249	0.0026 [0.0009, 0.0049]	0.618	0.1051 [0.0809, 0.1329]
2022	CJ–HJ	0.1786	0.0244 [0.0180, 0.0299]	0.299	0.5947 [0.5376, 0.6576]
	HP–HJ	0.8872	0.4995 [0.4768, 0.5246]	0.130	0.9575 [0.9388, 0.9741]
	HM–HJ	0.0376	0.0026 [0.0013, 0.0048]	0.581	0.1247 [0.0966, 0.1570]

Entries use each contrast's paired common-support population at the nominal review budget $q_0 = 0.10$; all Stage-II quantities use the same annual HJ reference cohort as Table 3. Interval estimates are episode-cluster percentile bootstrap intervals ($B = 2\,000$, seed fixed before the run). Stage-I displaced share is the number of reference shortlist positions replaced, divided by the total shortlist size across PRE and TURN. HM–HJ is interpreted as a dependence-only contrast subject to the weighted-marginal preservation audit reported in the text; L^{rec} is not intrinsically bounded by one. Absolute reference losses, reference denominators, and activation events are given in the Supplementary Material.

same stage-specific review capacity. Stage-I shortlist changes and retained consequences are assessed separately from Stage-II recovery activation, target agreement, and reference-objective loss.

Table 4 reports the annual results for the three principal comparisons. Stage-I L^{att} and Stage-II L^{rec} have different reference objects and denominators, so their magnitudes are not treated as a common performance scale.

The dependence-only contrast is small at both decision layers. Replacing the joint state coupling with the exact weighted-marginal coupling (HM–HJ) displaces 1.3–3.8% of shortlist positions with $L^{\text{att}} = 0.0006$–0.0026, and it leaves exact target agreement at $A_0 = 0.58$–0.73 with $L^{\text{rec}} = 0.085$–0.125. The Stage-I intervals exclude zero in every year, but the retained reference consequence is above 99.6% throughout. By contrast, collapsing the distribution to the weighted joint medoid (HP–HJ) displaces 84–89% of shortlist positions with $L^{\text{att}} = 0.41$–0.53; Stage-II exact agreement falls to $A_0 = 0.13$–0.24 and $L^{\text{rec}} = 0.90$–0.98. The paired Point-minus-Marginal increment attributes most of this loss to the discarded primitive marginal uncertainty rather than to cross-state alignment: the increment is

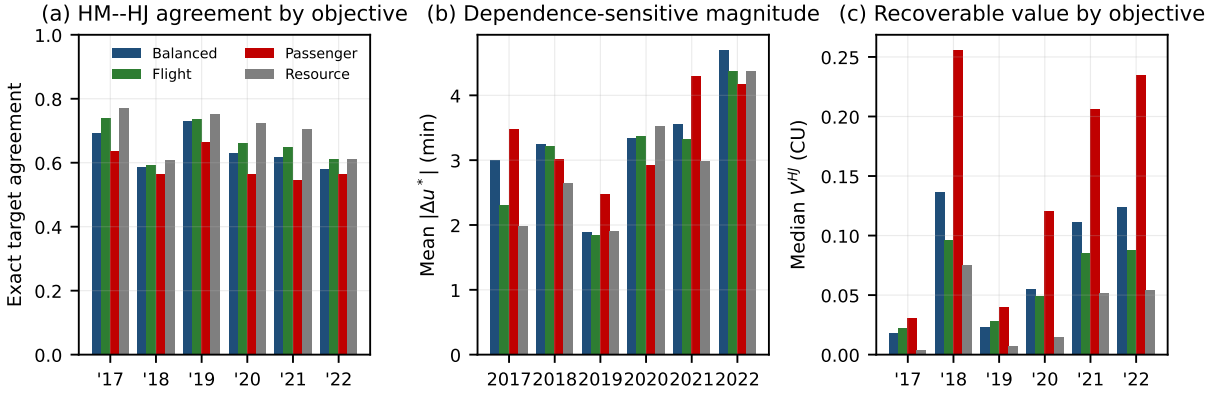


Figure 5: Decision sensitivity to joint-state dependence under alternative recovery objectives. Panel (a) reports exact HM–HJ target agreement within each prespecified objective; panel (b) reports the mean absolute target change $|\Delta u^*|$ across all cases (the median change is zero in every setting); panel (c) reports the median History+Joint recoverable value under each objective. The same annual reference cohorts are retained throughout.

$L^{\mathrm{att}} = 0.41$–0.53 and $L^{\mathrm{rec}} = 0.80$–0.89, and the residual Joint-alignment component is the same small HM–HJ quantity reported above. Restoring marginal uncertainty therefore recovers most of the observed Point-to-Joint difference, while additional joint alignment contributes little at either layer.

The temporal contrast is year-dependent. Under the Balanced objective the Current representation leaves L^{att} between 0.010 (2021) and 0.032 (2017) in the pre-pandemic years and 2021–2022, but $L^{\mathrm{att}} = 0.506$ in 2020 together with $A_0 = 0.086$ and $L^{\mathrm{rec}} = 0.998$: in the disruption year the current-state representation loses almost the entire reference recoverable value on the reference cohort, missing 425 of the 474 activation decisions. The Current and History models are trained with the same architecture size, normalization, protocol, and seed and differ only in the history-encoder mode, so this gap is not attributable to parameter capacity; it is nonetheless an implemented-configuration contrast rather than a pure information-content estimate.

A component-level audit explains why the dependence-only contrast can be small while the temporal and point contrasts are not. Under the exact marginal coupling, the Flight and Resource domain scores are numerically identical to their Joint counterparts for every audited candidate in all six years (maximum absolute difference below 10^{-13} CU), because those components are linear in the unresolved primitives. Only the Passenger domain differs, for 39–66% of candidates, and it does so through the threshold-event exposures whose probabilities depend on the coupling. Cross-state dependence therefore reaches the two decisions only through threshold-type consequences in this consequence system; this is a property of the specified objective, not a general statement about dependence.

5.5. Information Requirements under Alternative Recovery Objectives

The relative value of probabilistic dependence may change with the consequences that the recovery objective emphasizes. To examine this management question, we compare HM and HJ under four prespecified domain-weight settings: Balanced, Flight emphasis, Passenger emphasis, and Resource emphasis. Both targets are recomputed for each objective and evaluated under that objective's own Joint reference. The main comparison fixes the annual reference cohort independently of screening choice and domain weights, so preference changes do not also redefine the analyzed population.

Figure 5 reports the objective-by-distribution comparison. Its primary decision outcomes are exact target agreement, the fraction of cases whose optimal target is dependence-sensitive, and the mean absolute target change; fixed-target objective differences are reported separately.

Exact target agreement between the marginal and joint decisions ranges from $A_0 = 0.581$–0.730 under Balanced, 0.593–0.739 under Flight emphasis, 0.547–0.664 under Passenger emphasis, and 0.608–0.772 under Resource emphasis. Passenger emphasis is the least stable objective in every year except 2017, and its median recoverable value is the largest of the four settings (0.031–0.256 CU versus 0.004–0.075 CU under Resource emphasis). Resource emphasis changes the composition of the decision problem in the opposite direction: activation is lowest (0.631–0.835) and the zero-recoverable-value share highest (0.165–0.369), consistent with the invariance of the Resource component to the

local recovery action. No objective produces an invariant ranking across all six years: the largest annual agreement gap between any two objectives is reached in 2018 (Flight 0.739 versus Passenger 0.564).

Two mechanism results qualify these differences. First, the mean absolute target change under dependence is 1.8–4.7 minutes, but the median change is zero in all twenty-four year-objective settings, so most agreement is carried by cases where dependence does not move the optimum at all, with a tail of cases moving by 30–45 minutes. Second, the median decision margin is small (0.010–0.021 CU) relative to the objective perturbations introduced by reweighting domains, so objective-driven changes in target selection should be read together with the margin distribution rather than as evidence that one representation dominates.

5.6. Operating-stage and Regime Differences

The preceding analyses use annual evidence throughout rather than treating additional years as an appendix to a 2019 benchmark. This final subsection considers which relationships vary with the operating environment, without repeating the full annual metric tables. We compare PRE with TURN and examine annual contrasts in the pre-pandemic operating years (2017–2019), the pandemic disruption year (2020), and the recovery years (2021–2022). Differences are interpreted with their stage composition, eligibility, and evaluation scales; they do not isolate a causal effect of the pandemic or any particular external shock.

Screening disagreement between expected delay and consequence is lower in the pre-pandemic benchmark and in 2020 than in the recovery years, for both stages: the delay priority's L^{att} averages 0.133 (PRE) and 0.129 (TURN) in 2017–2019, falls to 0.101 and 0.107 in 2020, and rises to 0.169 in both stages across 2021–2022. The same ordering appears in the paired Stage-I displacement: delay replaces 39–46% of positions in the earlier years and 52–55% in 2021–2022. Loss magnitudes and population sizes move in opposite directions in 2020: the PRE candidate population contracts to 675 while the PRE screening loss reaches its six-year minimum.

The representation contrasts vary less by stage than by year. Under the Balanced objective, HM–HJ loses $L^{\mathrm{rec}} =$ 0.085–0.093 in 2017–2019, 0.091 in 2020, and 0.105–0.125 in 2021–2022, while HP–HJ loses 0.90–0.98 throughout; the ordering of the three contrasts is identical in all six years and both stages. The Current representation is the exception: its Stage-II loss is 0.31–0.70 in the pre-pandemic and recovery years and 0.998 in 2020, where its missed-activation count (425 of 474 reference-cohort cases) accounts for nearly the entire loss. Objective-level agreement is equally year-specific: HM–HJ target agreement under Balanced falls from 0.730 (2019) and 0.631 (2020) to 0.581 (2022), with Passenger emphasis remaining the least stable setting.

6. Discussion and Managerial Implications

Operational information does not have a fixed value independently of the managerial task it supports. The same operating state can inform which disruptions deserve early attention and whether a particular disruption still presents a worthwhile local-recovery opportunity. These decisions involve different consequences, alternatives, and comparison boundaries. Accordingly, the usefulness of a more detailed state representation must be assessed in relation to the recovery decision for which it is used. This section discusses the distinction between screening and local recovery, the decision-specific nature of information value, and the implications for airline operations control centers (AOCCs).

6.1. Screening Priority versus Recoverable Opportunity

Airline disruption management involves the allocation of both managerial attention and recovery effort, but these are not the same allocation problem. For an AOCC monitoring a large number of flights, initial screening identifies disruptions whose operational consequences warrant further diagnosis. Once a flight chain enters detailed review, the relevant question becomes how much recovery opportunity remains and whether the expected improvement justifies additional effort. Screening compares priorities across flight chains; local recovery compares feasible recovery targets within a given chain.

Consequently, consequence severity cannot be treated as a direct proxy for recoverable value. High exposure may reflect predecessor delay, passenger volumes, downstream connections, or other operating conditions, not all of which can be improved through the available local adjustment. A flight chain may remain consequential even when little turnaround headroom remains or when further off-block advancement no longer changes the principal consequences. Conversely, a chain with more moderate baseline exposure may offer a comparatively accessible local improvement. The relevant distinction is between the consequences at risk and the consequences that can still be improved under the current operating state.

The two decision layers therefore allocate different forms of scarce managerial capacity. Consequence-based screening seeks to direct limited detailed review toward important disruptions; local-recovery targeting evaluates the remaining opportunity against the effort required to realize it. A screened chain for which the preferred target is zero does not, by itself, imply that the screening decision was misplaced: detailed diagnosis may establish that substantial exposure coexists with limited additional recovery options.

This distinction also clarifies the choice of screening criterion. When detailed review is intended to prioritize consequential disruptions, baseline consequence exposure is the direct evaluation object. When the principal objective is to identify feasible improvement opportunities, recoverable-value coverage becomes an additional consideration. The performance of a shortlist must therefore be interpreted against its intended managerial purpose rather than judged solely by the number of subsequently selected positive recovery targets.

6.2. Decision-Specific Value of Operational Information

The managerial value of operating information depends not only on the state detail it preserves but also on whether that detail changes the comparison among the alternatives relevant to the current decision. Under fixed-budget screening, the critical comparison concerns flight chains near the shortlist cutoff. Under local recovery, it concerns the expected consequence–effort trade-off among feasible targets. Distinct state predictions can therefore support the same decision, while relatively small changes in the evaluation of closely competing alternatives can alter the preferred choice.

This helps explain why predictive quality and decision value need not move together. A richer operating history may improve the description of state evolution without changing the cross-chain screening order. Equally, information differences that are modest in aggregate predictive terms may affect the comparison between zero and positive recovery or between two positive targets. Information evaluation should therefore consider both whether the chosen alternative changes and the reference-objective consequence of that change. Prediction error, overall ranking agreement, and target agreement each describe only part of this relationship.

The same reasoning applies to probabilistic representations. A joint distribution retains dependence among unresolved operating states, but statistical completeness does not imply that all recovery decisions require this structure. When the expected consequence of each fixed alternative is fully determined by separable marginal contributions, preserving the primitive marginals can suffice. In contrast, jointly determined thresholds, state-dependent headroom, and other nonseparable relationships can make cross-state dependence relevant to alternative evaluation. Even in these settings, a difference in expected consequence changes the preferred decision only when it alters the relative ordering of feasible alternatives.

Managerial objectives can further modify these information requirements. Airline recovery balances flight operations, passenger exposure, and operating resources, and changing their relative weights changes how individual consequences enter the objective. Preference weights and objective structure nevertheless play distinct roles. Increasing the emphasis on a consequence domain does not necessarily increase sensitivity to joint dependence: the domain must actually involve relevant cross-state relationships for that channel to affect the recovery comparison. Otherwise, a weight change may primarily rescale the trade-off between other improvable consequences and recovery effort.

Decision sufficiency is thus a relationship among the operating state, the active managerial objective, and the decision boundary, rather than an intrinsic property of a forecasting representation. As operational events become known, remaining uncertainty and recovery opportunities may change together. The same representation may therefore be adequate at one decision point and insufficient at another, depending on which alternatives remain feasible and how closely they are evaluated.

6.3. Implications for AOCC Decision Support

These distinctions have implications for how AOCCs organize information for rolling disruption management. A control center must monitor many evolving flight states while balancing analytical timeliness against the depth of evaluation. Different disruptions and decision tasks need not require identical state models or probabilistic detail. Decision support can therefore be organized around the information needed for the current managerial choice rather than around a uniform objective of maximizing representational richness.

At the initial screening stage, timely and interpretable delay-based signals provide a compact basis for comparing many disrupted flight chains. Multidimensional exposure information can become especially relevant when flights with similar delays differ in passenger, downstream, or resource consequences, or when candidate scores are close to the

active review cutoff. The operational objective is to distinguish which cases merit more detailed attention, not to apply the most elaborate representation indiscriminately to every flight.

Once a chain enters detailed diagnosis, attention shifts from baseline exposure to remaining recovery opportunity. The relevant questions concern which consequences are responsive to a local adjustment, how much effective headroom remains, and whether the improvement is sufficient to justify additional effort. At this layer, historical trajectories and unresolved-state uncertainty may affect recovery activation or target intensity. Their usefulness should nevertheless be judged by the recovery alternatives they distinguish, not by the completeness of the resulting forecast alone.

These observations motivate *decision-matched information provision*: the analytical detail supplied to an AOCC should correspond to the requirements of the recovery task. Initial screening emphasizes timeliness and cross-flight comparability; detailed diagnosis adds multidimensional exposure and headroom; local-recovery evaluation focuses on uncertainty, feasible alternatives, and the consequence–effort trade-off. This is not a fixed progression from delay to consequences, history, and joint distributions. A simpler representation may already preserve the decision-relevant distinctions for a given objective and operating state.

Analytical depth can itself be regarded as a constrained decision-support resource. Flights close to a screening boundary, or recovery targets with small objective gaps and material cross-state coupling, may be natural candidates for more detailed evaluation. Where one alternative is clearly preferred, further statistical refinement may have little influence on the supported decision. Decision boundaries can thus inform where additional analytical effort is potentially useful, not merely which recovery alternative is preferred.

Decision-matched information provision is a design principle derived from the decision structure and the controlled comparisons, rather than a validated policy for optimally sequencing information acquisition. Operational deployment would additionally require evidence on analytical latency, computational cost, controller review capacity, and the performance of executable recovery actions.

6.4. Applicability and Future Research

The evaluation design draws on publicly available U.S. airline operating records from 2017 to 2022, covering pre-pandemic operations, the COVID-19 disruption, and the subsequent recovery period. These distinct environments permit examination of whether decision relationships persist under changing operating conditions. Year-to-year differences may nonetheless reflect concurrent changes in traffic volume, network structure, recovery opportunities, and passenger exposure. Cross-year comparisons characterize empirical stability and conditionality; they do not isolate the causal effect of the pandemic or any individual external shock.

Public operating records support the reconstruction of flight execution, aircraft continuity, and several historical consequence-exposure proxies. Actual AOCC recovery also involves carrier-internal information on crew legality, gate and ground-handling competition, maintenance restrictions, reserve aircraft, real-time passenger reaccommodation, and operational judgments. Such information may affect both exposure and remaining recovery feasibility. Access to carrier-internal operating and decision records would allow the incremental decision value of these signals to be assessed under more complete operational constraints.

The local-recovery decision is represented by an effective target constrained by modeled time headroom. Implementing a recovery decision would additionally require translating that target into feasible resource actions. Incorporating aircraft swaps, crew adjustments, gate coordination, and shared turnaround resources would move the framework toward resource-explicit recovery optimization. Such decisions may also introduce stronger cross-flight and cross-resource dependencies, changing the role of joint-state information in feasibility and recourse evaluation.

Finally, the multidimensional consequence criterion uses transparent normalization and normative domain weights. Carrier-specific trade-offs among operational reliability, passenger service, and resource expenditure will depend on network structure, costs, service commitments, and operating policies. Linking the framework to observed carrier costs, managerial preferences, and executed recovery outcomes would permit evaluation of organization-specific information requirements and the operational effectiveness of decision-matched information provision.

7. Conclusion

This study examines when operational information is sufficient for decisions in rolling airline disruption recovery. Using a common multidimensional consequence criterion and two distinct decision tasks—fixed-budget recovery screening and effective local-recovery targeting—the empirical design evaluates delay-based priorities, temporal information, and probabilistic state representations across U.S. airline operations in 2017–2022. The central question

is not whether a representation retains more operating-state detail, but whether that detail changes the allocation of limited managerial attention or the selection of a worthwhile local-recovery target.

The empirical conclusions will be organized around three related findings. First, the six-year screening comparisons establish *[E1 finding pending: when delay-based screening does or does not preserve consequence-based shortlists and their reference consequence coverage]*. Second, the full eligible recovery cohorts show *[E2 finding pending: the observed relationship between baseline consequence priority and remaining recoverable opportunity]*; these are distinct managerial criteria rather than interchangeable measures. Third, the temporal and distributional comparisons, together with objective-specific and operating-regime analyses, indicate *[E2–E4 finding pending: which information distinctions alter decisions, under which objectives and operating conditions]*. These findings must be interpreted through the corresponding decision alternatives and reference losses, rather than inferred from predictive accuracy or representational richness alone.

The managerial implication is *decision-matched information provision*: the analytical detail supplied to an airline operations control center should correspond to the decision it supports. Initial screening, detailed consequence assessment, and local-recovery targeting need not require the same information depth at every operating epoch. Translating this principle into executable airline recovery would require carrier-internal operating information, resource-explicit recovery actions, and evaluation against realized operational outcomes.

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